

Heteroskedasticity

The homoskedasticity assumption, introduced in Chapter 3 for multiple regression, states that the variance of the unobserved error, u , conditional on the explanatory variables, is constant. Homoskedasticity fails whenever the variance of the unobserved factors changes across different segments of the population, where the segments are determined by the different values of the explanatory variables. For example, in a savings equation, heteroskedasticity is present if the variance of the unobserved factors affecting savings increases with income.

In Chapters 4 and 5, we saw that homoskedasticity is needed to justify the usual t tests, F tests, and confidence intervals for OLS estimation of the linear regression model, even with large sample sizes. In this chapter, we discuss the available remedies when heteroskedasticity occurs, and we also show how to test for its presence. We begin by briefly reviewing the consequences of heteroskedasticity for ordinary least squares estimation.

8.1 Consequences of Heteroskedasticity for OLS

Consider again the multiple linear regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u. \quad [8.1]$$

In Chapter 3, we proved unbiasedness of the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$ under the first four Gauss-Markov assumptions, MLR.1 through MLR.4. In Chapter 5, we showed that the same four assumptions imply consistency of OLS. The homoskedasticity assumption MLR.5, stated in terms of the error variance as $\text{Var}(u|x_1, x_2, \dots, x_k) = \sigma^2$, played no role in showing whether OLS was unbiased or consistent. It is important to remember that heteroskedasticity does not cause bias or inconsistency in the OLS estimators of the β_j , whereas something like omitting an important variable would have this effect.

The interpretation of our goodness-of-fit measures, R^2 and $\bar{R}^2$, is also unaffected by the presence of heteroskedasticity. Why? Recall from Section 6.3 that the usual R -squared and the adjusted R -squared are different ways of estimating the population R -squared, which is simply $1 - \sigma_u^2/\sigma_y^2$, where σ_u^2 is the population error variance and σ_y^2 is the

population variance of y . The key point is that because both variances in the population R -squared are unconditional variances, the population R -squared is unaffected by the presence of heteroskedasticity in $\text{Var}(u|x_1, \dots, x_k)$. Further, SSR/n consistently estimates σ_u^2 , and SST/n consistently estimates σ_y^2 , whether or not $\text{Var}(u|x_1, \dots, x_k)$ is constant. The same is true when we use the degrees of freedom adjustments. Therefore, R^2 and $\bar{R}^2$ are both consistent estimators of the population R -squared whether or not the homoskedasticity assumption holds.

If heteroskedasticity does not cause bias or inconsistency in the OLS estimators, why did we introduce it as one of the Gauss-Markov assumptions? Recall from Chapter 3 that the estimators of the *variances*, $\text{Var}(\hat{\beta}_j)$, are biased without the homoskedasticity assumption. Since the OLS standard errors are based directly on these variances, they are no longer valid for constructing confidence intervals and t statistics. The usual OLS t statistics do not have t distributions in the presence of heteroskedasticity, and the problem is not resolved by using large sample sizes. We will see this explicitly for the simple regression case in the next section, where we derive the variance of the OLS slope estimator under heteroskedasticity and propose a valid estimator in the presence of heteroskedasticity. Similarly, F statistics are no longer F distributed, and the LM statistic no longer has an asymptotic chi-square distribution. In summary, the statistics we used to test hypotheses under the Gauss-Markov assumptions are not valid in the presence of heteroskedasticity.

We also know that the Gauss-Markov Theorem, which says that OLS is best linear unbiased, relies crucially on the homoskedasticity assumption. If $\text{Var}(u|\mathbf{x})$ is not constant, OLS is no longer BLUE. In addition, OLS is no longer asymptotically efficient in the class of estimators described in Theorem 5.3. As we will see in Section 8.4, it is possible to find estimators that are more efficient than OLS in the presence of heteroskedasticity (although it requires knowing the form of the heteroskedasticity). With relatively large sample sizes, it might not be so important to obtain an efficient estimator. In the next section, we show how the usual OLS test statistics can be modified so that they are valid, at least asymptotically.

8.2 Heteroskedasticity-Robust Inference after OLS Estimation

Because testing hypotheses is such an important component of any econometric analysis and the usual OLS inference is generally faulty in the presence of heteroskedasticity, we must decide if we should entirely abandon OLS. Fortunately, OLS is still useful. In the last two decades, econometricians have learned how to adjust standard errors and t , F , and LM statistics so that they are valid in the presence of **heteroskedasticity of unknown form**. This is very convenient because it means we can report new statistics that work regardless of the kind of heteroskedasticity present in the population. The methods in this section are known as *heteroskedasticity-robust* procedures because they are valid—at least in large samples—whether or not the errors have constant variance, and we do not need to know which is the case.

We begin by sketching how the variances, $\text{Var}(\hat{\beta}_j)$, can be estimated in the presence of heteroskedasticity. A careful derivation of the theory is well beyond the scope of this text, but the application of heteroskedasticity-robust methods is very easy now because many statistics and econometrics packages compute these statistics as an option.

First, consider the model with a single independent variable, where we include an i subscript for emphasis:

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

We assume throughout that the first four Gauss-Markov assumptions hold. If the errors contain heteroskedasticity, then

$$\text{Var}(u_i|x_i) = \sigma_i^2,$$

where we put an i subscript on σ^2 to indicate that the variance of the error depends upon the particular value of x_i .

Write the OLS estimator as

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Under Assumptions MLR.1 through MLR.4 (that is, without the homoskedasticity assumption), and conditioning on the values x_i in the sample, we can use the same arguments from Chapter 2 to show that

$$\text{Var}(\hat{\beta}_1) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma_i^2}{\text{SST}_x^2}, \quad [8.2]$$

where $\text{SST}_x = \sum_{i=1}^n (x_i - \bar{x})^2$ is the total sum of squares of the x_i . When $\sigma_i^2 = \sigma^2$ for all i , this formula reduces to the usual form, σ^2/SST_x . Equation (8.2) explicitly shows that, for the simple regression case, the variance formula derived under homoskedasticity is no longer valid when heteroskedasticity is present.

Since the standard error of $\hat{\beta}_1$ is based directly on estimating $\text{Var}(\hat{\beta}_1)$, we need a way to estimate equation (8.2) when heteroskedasticity is present. White (1980) showed how this can be done. Let $\hat{u}_i$ denote the OLS residuals from the initial regression of y on x . Then, a valid estimator of $\text{Var}(\hat{\beta}_1)$, for heteroskedasticity of *any* form (including homoskedasticity), is

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2 \hat{u}_i^2}{\text{SST}_x^2}, \quad [8.3]$$

which is easily computed from the data after the OLS regression.

In what sense is (8.3) a valid estimator of $\text{Var}(\hat{\beta}_1)$? This is pretty subtle. Briefly, it can be shown that when equation (8.3) is multiplied by the sample size n , it converges in probability to $E[(x_i - \mu_x)^2 u_i^2]/(\sigma_x^2)^2$, which is the probability limit of n times (8.2). Ultimately, this is what is necessary for justifying the use of standard errors to construct confidence intervals and t statistics. The law of large numbers and the central limit theorem play

key roles in establishing these convergences. You can refer to White's original paper for details, but that paper is quite technical. See also Wooldridge (2010, Chapter 4).

A similar formula works in the general multiple regression model

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u.$$

It can be shown that a valid estimator of $\text{Var}(\hat{\beta}_j)$, under Assumptions MLR.1 through MLR.4, is

$$\widehat{\text{Var}}(\hat{\beta}_j) = \frac{\sum_{i=1}^n \hat{r}_{ij}^2 \hat{u}_i^2}{\text{SSR}_j}, \quad [8.4]$$

where $\hat{r}_{ij}$ denotes the i^{th} residual from regressing x_j on all other independent variables, and SSR_j is the sum of squared residuals from this regression (see Section 3.2 for the partialling out representation of the OLS estimates). The square root of the quantity in (8.4) is called the **heteroskedasticity-robust standard error** for $\hat{\beta}_j$. In econometrics, these robust standard errors are usually attributed to White (1980). Earlier works in statistics, notably those by Eicker (1967) and Huber (1967), pointed to the possibility of obtaining such robust standard errors. In applied work, these are sometimes called *White*, *Huber*, or *Eicker standard errors* (or some hyphenated combination of these names). We will just refer to them as *heteroskedasticity-robust standard errors*, or even just *robust standard errors* when the context is clear.

Sometimes, as a degrees of freedom correction, (8.4) is multiplied by $n/(n - k - 1)$ before taking the square root. The reasoning for this adjustment is that, if the squared OLS residuals $\hat{u}_i^2$ were the same for all observations i —the strongest possible form of homoskedasticity in a sample—we would get the usual OLS standard errors. Other modifications of (8.4) are studied in MacKinnon and White (1985). Since all forms have only asymptotic justification and they are asymptotically equivalent, no form is uniformly preferred above all others. Typically, we use whatever form is computed by the regression package at hand.

Once heteroskedasticity-robust standard errors are obtained, it is simple to construct a **heteroskedasticity-robust t statistic**. Recall that the general form of the t statistic is

$$t = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}. \quad [8.5]$$

Because we are still using the OLS estimates and we have chosen the hypothesized value ahead of time, the only difference between the usual OLS t statistic and the heteroskedasticity-robust t statistic is in how the standard error in the denominator is computed.

The term SSR_j in equation (8.4) can be replaced with $\text{SST}_j(1 - R_j^2)$, where SST_j is the total sum of squares of x_j and R_j^2 is the usual R -squared from regressing x_j on all other explanatory variables. [We implicitly used this equivalence in deriving equation (3.51).] Consequently, little sample variation in x_j , or a strong linear relationship between x_j and the other explanatory variables—that is, multicollinearity—can cause the heteroskedasticity-robust standard errors to be large. We discussed these issues with the usual OLS standard errors in Section 3.4.

EXAMPLE 8.1

LOG WAGE EQUATION WITH HETEROSKEDASTICITY-ROBUST STANDARD ERRORS

We estimate the model in Example 7.6, but we report the heteroskedasticity-robust standard errors along with the usual OLS standard errors. Some of the estimates are reported to more digits so that we can compare the usual standard errors with the heteroskedasticity-robust standard errors:

$$\begin{aligned} \widehat{\log(\text{wage})} = & .321 + .213 \text{ marrmale} - .198 \text{ marrfem} - .110 \text{ singfem} \\ & (.100) \quad (.055) \quad (.058) \quad (.056) \\ & [.109] \quad [.057] \quad [.058] \quad [.057] \\ & + .0789 \text{ educ} + .0268 \text{ exper} - .00054 \text{ exper}^2 \\ & (.0067) \quad (.0055) \quad (.00011) \\ & [.0074] \quad [.0051] \quad [.00011] \\ & + .0291 \text{ tenure} - .00053 \text{ tenure}^2 \\ & (.0068) \quad (.00023) \\ & [.0069] \quad [.00024] \\ n = 526, R^2 = .461. \end{aligned} \tag{8.6}$$

The usual OLS standard errors are in parentheses, (), below the corresponding OLS estimate, and the heteroskedasticity-robust standard errors are in brackets, []. The numbers in brackets are the only new things, since the equation is still estimated by OLS.

Several things are apparent from equation (8.6). First, in this particular application, any variable that was statistically significant using the usual t statistic is still statistically significant using the heteroskedasticity-robust t statistic. This occurs because the two sets of standard errors are not very different. (The associated p -values will differ slightly because the robust t statistics are not identical to the usual, nonrobust t statistics.) The largest relative change in standard errors is for the coefficient on *educ*: the usual standard error is .0067, and the robust standard error is .0074. Still, the robust standard error implies a robust t statistic above 10.

Equation (8.6) also shows that the robust standard errors can be either larger or smaller than the usual standard errors. For example, the robust standard error on *exper* is .0051, whereas the usual standard error is .0055. We do not know which will be larger ahead of time. As an empirical matter, the robust standard errors are often found to be larger than the usual standard errors.

Before leaving this example, we must emphasize that we do not know, at this point, whether heteroskedasticity is even present in the population model underlying equation (8.6). All we have done is report, along with the usual standard errors, those that are valid (asymptotically) whether or not heteroskedasticity is present. We can see that no important conclusions are overturned by using the robust standard errors in this example. This often happens in applied work, but in other cases, the differences between the usual and robust standard errors are much larger. As an example of where the differences are substantial, see Computer Exercise C2.

At this point, you may be asking the following question: If the heteroskedasticity-robust standard errors are valid more often than the usual OLS standard errors, why do we bother with the usual standard errors at all? This is a sensible question. One reason the

usual standard errors are still used in cross-sectional work is that, if the homoskedasticity assumption holds and the errors are normally distributed, then the usual t statistics have *exact* t distributions, regardless of the sample size (see Chapter 4). The robust standard errors and robust t statistics are justified only as the sample size becomes large, even if the CLM assumptions are true. With small sample sizes, the robust t statistics can have distributions that are not very close to the t distribution, and that could throw off our inference.

In large sample sizes, we can make a case for always reporting only the heteroskedasticity-robust standard errors in cross-sectional applications, and this practice is being followed more and more in applied work. It is also common to report both standard errors, as in equation (8.6), so that a reader can determine whether any conclusions are sensitive to the standard error in use.

It is also possible to obtain F and LM statistics that are robust to heteroskedasticity of an unknown, arbitrary form. The **heteroskedasticity-robust F statistic** (or a simple transformation of it) is also called a *heteroskedasticity-robust Wald statistic*. A general treatment of the Wald statistic requires matrix algebra and is sketched in Appendix E; see Wooldridge (2010, Chapter 4) for a more detailed treatment. Nevertheless, using heteroskedasticity-robust statistics for multiple exclusion restrictions is straightforward because many econometrics packages now compute such statistics routinely.

EXAMPLE 8.2**HETEROSKEDASTICITY-ROBUST F STATISTIC**

Using the data for the spring semester in GPA3.RAW, we estimate the following equation:

$$\begin{aligned} \widehat{cumgpa} = & 1.47 + .00114 sat - .00857 hspec + .00250 tothrs \\ & (.23) \quad (.00018) \quad (.00124) \quad (.00073) \\ & [.22] \quad [.00019] \quad [.00140] \quad [.00073] \\ & + .303 female - .128 black - .059 white \\ & (.059) \quad (.147) \quad (.141) \\ & [.059] \quad [.118] \quad [.110] \\ n = 366, R^2 = .4006, \bar{R}^2 = .3905. \end{aligned} \quad [8.7]$$

Again, the differences between the usual standard errors and the heteroskedasticity-robust standard errors are not very big, and use of the robust t statistics does not change the statistical significance of any independent variable. Joint significance tests are not much affected either. Suppose we wish to test the null hypothesis that, after the other factors are controlled for, there are no differences in *cumgpa* by race. This is stated as $H_0: \beta_{black} = 0, \beta_{white} = 0$. The usual F statistic is easily obtained, once we have the R -squared from the restricted model; this turns out to be .3983. The F statistic is then $[(.4006 - .3983)/(1 - .4006)](359/2) \approx .69$. If heteroskedasticity is present, this version of the test is invalid. The heteroskedasticity-robust version has no simple form, but it can be computed using certain statistical packages. The value of the heteroskedasticity-robust F statistic turns out to be .75, which differs only slightly from the nonrobust version. The p -value for the robust test is .474, which is not close to standard significance levels. We fail to reject the null hypothesis using either test.

Because the usual sum-of-squared residuals form of the F statistic is not valid under heteroskedasticity, it can be important when computing a Chow test to use the form that includes a full set of interactions between the group dummy variable and the other

explanatory variables. This can be tedious but is necessary for obtaining heteroskedasticity-robust F statistics using popular regression packages. See Computer Exercise C14 for an example.

Computing Heteroskedasticity-Robust LM Tests

EXPLORING FURTHER 8.1

Evaluate the following statement: The heteroskedasticity-robust standard errors are always bigger than the usual standard errors.

Not all regression packages compute F statistics that are robust to heteroskedasticity. Therefore, it is sometimes convenient to have a way of obtaining a test of multiple exclusion restrictions that is robust to heteroskedasticity and does not require a particular kind of econometric

software. It turns out that a **heteroskedasticity-robust LM statistic** is easily obtained using virtually any regression package.

To illustrate computation of the robust LM statistic, consider the model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + u,$$

and suppose we would like to test $H_0: \beta_4 = 0, \beta_5 = 0$. To obtain the usual LM statistic, we would first estimate the restricted model (that is, the model without x_4 and x_5) to obtain the residuals, $\tilde{u}$. Then, we would regress $\tilde{u}$ on all of the independent variables and the $LM = n \cdot R_{\tilde{u}}^2$ where $R_{\tilde{u}}^2$ is the usual R -squared from this regression.

Obtaining a version that is robust to heteroskedasticity requires more work. One way to compute the statistic requires only OLS regressions. We need the residuals, say, $\tilde{r}_1$, from the regression of x_4 on x_1, x_2, x_3 . Also, we need the residuals, say, $\tilde{r}_2$, from the regression of x_5 on x_1, x_2, x_3 . Thus, we regress each of the independent variables excluded under the null on all of the included independent variables. We keep the residuals each time. The final step appears odd, but it is, after all, just a computational device. Run the regression of

$$1 \text{ on } \tilde{r}_1 \tilde{u}, \tilde{r}_2 \tilde{u}, \quad [8.8]$$

without an intercept. Yes, we actually define a dependent variable equal to the value one for all observations. We regress this onto the products $\tilde{r}_1 \tilde{u}$ and $\tilde{r}_2 \tilde{u}$. The robust LM statistic turns out to be $n - SSR_1$, where SSR_1 is just the usual sum of squared residuals from regression (8.8).

The reason this works is somewhat technical. Basically, this is doing for the LM test what the robust standard errors do for the t test. [See Wooldridge (1991b) or Davidson and MacKinnon (1993) for a more detailed discussion.]

We now summarize the computation of the heteroskedasticity-robust LM statistic in the general case.

A Heteroskedasticity-Robust LM Statistic:

1. Obtain the residuals $\tilde{u}$ from the restricted model.
2. Regress each of the independent variables excluded under the null on all of the included independent variables; if there are q excluded variables, this leads to q sets of residuals $(\tilde{r}_1, \tilde{r}_2, \dots, \tilde{r}_q)$.
3. Find the products between each $\tilde{r}_j$ and $\tilde{u}$ (for all observations).

4. Run the regression of 1 on $\tilde{r}_1\tilde{u}, \tilde{r}_2\tilde{u}, \dots, \tilde{r}_q\tilde{u}$, without an intercept. The heteroskedasticity-robust LM statistic is $n - SSR_1$, where SSR_1 is just the usual sum of squared residuals from this final regression. Under H_0 , LM is distributed approximately as χ_q^2 .

Once the robust LM statistic is obtained, the rejection rule and computation of p -values are the same as for the usual LM statistic in Section 5.2.

EXAMPLE 8.3**HETEROSKEDASTICITY-ROBUST LM STATISTIC**

We use the data in CRIME1.RAW to test whether the average sentence length served for past convictions affects the number of arrests in the current year (1986). The estimated model is

$$\begin{aligned} \widehat{narr86} = & .567 - .136 pcnv + .0178 avg\text{sen} - .00052 avg\text{sen}^2 \\ & (.036) \quad (.040) \quad (.0097) \quad (.00030) \\ & [.040] \quad [.034] \quad [.0101] \quad [.00021] \\ & - .0394 ptime86 - .0505 qemp86 - .00148 inc86 \\ & (.0087) \quad (.0144) \quad (.00034) \quad [8.9] \\ & [.0062] \quad [.0142] \quad [.00023] \\ & + .325 black + .193 hispan \\ & (.045) \quad (.040) \\ & [.058] \quad [.040] \\ n = & 2,725, R^2 = .0728. \end{aligned}$$

In this example, there are more substantial differences between some of the usual standard errors and the robust standard errors. For example, the usual t statistic on $avg\text{sen}^2$ is about -1.73 , while the robust t statistic is about -2.48 . Thus, $avg\text{sen}^2$ is more significant using the robust standard error.

The effect of $avg\text{sen}$ on $narr86$ is somewhat difficult to reconcile. Because the relationship is quadratic, we can figure out where $avg\text{sen}$ has a positive effect on $narr86$ and where the effect becomes negative. The turning point is $.0178/[2(.00052)] \approx 17.12$; recall that this is measured in months. Literally, this means that $narr86$ is positively related to $avg\text{sen}$ when $avg\text{sen}$ is less than 17 months; then $avg\text{sen}$ has the expected deterrent effect after 17 months.

To see whether average sentence length has a statistically significant effect on $narr86$, we must test the joint hypothesis $H_0: \beta_{avg\text{sen}} = 0, \beta_{avg\text{sen}^2} = 0$. Using the usual LM statistic (see Section 5.2), we obtain $LM = 3.54$; in a chi-square distribution with two df , this yields a p -value = .170. Thus, we do not reject H_0 at even the 15% level. The heteroskedasticity-robust LM statistic is $LM = 4.00$ (rounded to two decimal places), with a p -value = .135. This is still not very strong evidence against H_0 ; $avg\text{sen}$ does not appear to have a strong effect on $narr86$. [Incidentally, when $avg\text{sen}$ appears alone in (8.9), that is, without the quadratic term, its usual t statistic is .658, and its robust t statistic is .592.]

8.3 Testing for Heteroskedasticity

The heteroskedasticity-robust standard errors provide a simple method for computing t statistics that are asymptotically t distributed whether or not heteroskedasticity is present. We have also seen that heteroskedasticity-robust F and LM statistics are available.

Implementing these tests does not require knowing whether or not heteroskedasticity is present. Nevertheless, there are still some good reasons for having simple tests that can detect its presence. First, as we mentioned in the previous section, the usual t statistics have exact t distributions under the classical linear model assumptions. For this reason, many economists still prefer to see the usual OLS standard errors and test statistics reported, unless there is evidence of heteroskedasticity. Second, if heteroskedasticity is present, the OLS estimator is no longer the best linear unbiased estimator. As we will see in Section 8.4, it is possible to obtain a better estimator than OLS when the form of heteroskedasticity is known.

Many tests for heteroskedasticity have been suggested over the years. Some of them, while having the ability to detect heteroskedasticity, do not directly test the assumption that the variance of the error does not depend upon the independent variables. We will restrict ourselves to more modern tests, which detect the kind of heteroskedasticity that invalidates the usual OLS statistics. This also has the benefit of putting all tests in the same framework.

As usual, we start with the linear model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u, \quad [8.10]$$

where Assumptions MLR.1 through MLR.4 are maintained in this section. In particular, we assume that $E(u|x_1, x_2, \dots, x_k) = 0$, so that OLS is unbiased and consistent.

We take the null hypothesis to be that Assumption MLR.5 is true:

$$H_0: \text{Var}(u|x_1, x_2, \dots, x_k) = \sigma^2. \quad [8.11]$$

That is, we assume that the ideal assumption of homoskedasticity holds, and we require the data to tell us otherwise. If we cannot reject (8.11) at a sufficiently small significance level, we usually conclude that heteroskedasticity is not a problem. However, remember that we never accept H_0 ; we simply fail to reject it.

Because we are assuming that u has a zero conditional expectation, $\text{Var}(u|\mathbf{x}) = E(u^2|\mathbf{x})$, and so the null hypothesis of homoskedasticity is equivalent to

$$H_0: E(u^2|x_1, x_2, \dots, x_k) = E(u^2) = \sigma^2.$$

This shows that, in order to test for violation of the homoskedasticity assumption, we want to test whether u^2 is related (in expected value) to one or more of the explanatory variables. If H_0 is false, the expected value of u^2 , given the independent variables, can be virtually any function of the x_j . A simple approach is to assume a linear function:

$$u^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_k x_k + v, \quad [8.12]$$

where v is an error term with mean zero given the x_j . Pay close attention to the dependent variable in this equation: it is the *square* of the error in the original regression equation, (8.10). The null hypothesis of homoskedasticity is

$$H_0: \delta_1 = \delta_2 = \dots = \delta_k = 0. \quad [8.13]$$

Under the null hypothesis, it is often reasonable to assume that the error in (8.12), v , is independent of $x_1, x_2, \dots, x_k$. Then, we know from Section 5.2 that either the F or LM statistics for the overall significance of the independent variables in explaining u^2 can be used to test (8.13). Both statistics would have asymptotic justification, even though u^2 cannot

be normally distributed. (For example, if u is normally distributed, then u^2/σ^2 is distributed as χ_1^2 .) If we could observe the u^2 in the sample, then we could easily compute this statistic by running the OLS regression of u^2 on $x_1, x_2, \dots, x_k$, using all n observations.

As we have emphasized before, we never know the actual errors in the population model, but we do have estimates of them: the OLS residual, $\hat{u}_i$, is an estimate of the error u_i for observation i . Thus, we can estimate the equation

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_k x_k + \text{error} \quad [8.14]$$

and compute the F or LM statistics for the joint significance of $x_1, \dots, x_k$. It turns out that using the OLS residuals in place of the errors does not affect the large sample distribution of the F or LM statistics, although showing this is pretty complicated.

The F and LM statistics both depend on the R -squared from regression (8.14); call this R_u^2 to distinguish it from the R -squared in estimating equation (8.10). Then, the F statistic is

$$F = \frac{R_u^2/k}{(1 - R_u^2)/(n - k - 1)}, \quad [8.15]$$

where k is the number of regressors in (8.14); this is the same number of independent variables in (8.10). Computing (8.15) by hand is rarely necessary, because most regression packages automatically compute the F statistic for overall significance of a regression. This F statistic has (approximately) an $F_{k, n-k-1}$ distribution under the null hypothesis of homoskedasticity.

The LM statistic for heteroskedasticity is just the sample size times the R -squared from (8.14):

$$LM = n \cdot R_u^2. \quad [8.16]$$

Under the null hypothesis, LM is distributed asymptotically as χ_k^2 . This is also very easy to obtain after running regression (8.14).

The LM version of the test is typically called the **Breusch-Pagan test for heteroskedasticity (BP test)**. Breusch and Pagan (1979) suggested a different form of the test that assumes the errors are normally distributed. Koenker (1981) suggested the form of the LM statistic in (8.16), and it is generally preferred due to its greater applicability.

We summarize the steps for testing for heteroskedasticity using the BP test:

The Breusch-Pagan Test for Heteroskedasticity:

1. Estimate the model (8.10) by OLS, as usual. Obtain the squared OLS residuals, $\hat{u}^2$ (one for each observation).
2. Run the regression in (8.14). Keep the R -squared from this regression, R_u^2 .
3. Form either the F statistic or the LM statistic and compute the p -value (using the $F_{k, n-k-1}$ distribution in the former case and the χ_k^2 distribution in the latter case). If the p -value is sufficiently small, that is, below the chosen significance level, then we reject the null hypothesis of homoskedasticity.

If the BP test results in a small enough p -value, some corrective measure should be taken. One possibility is to just use the heteroskedasticity-robust standard errors and test statistics discussed in the previous section. Another possibility is discussed in Section 8.4.

EXAMPLE 8.4

HETEROSKEDASTICITY IN HOUSING PRICE EQUATIONS

We use the data in HPRICE1.RAW to test for heteroskedasticity in a simple housing price equation. The estimated equation using the levels of all variables is

$$\widehat{\text{price}} = -21.77 + .00207 \text{ lotsize} + .123 \text{ sqrft} + 13.85 \text{ bdrms} \\ (29.48) \quad (.00064) \quad (.013) \quad (9.01) \quad [8.17] \\ n = 88, R^2 = .672.$$

This equation tells us *nothing* about whether the error in the population model is heteroskedastic. We need to regress the squared OLS residuals on the independent variables. The R -squared from the regression of $\hat{u}^2$ on *lotsize*, *sqrft*, and *bdrms* is $R_u^2 = .1601$. With $n = 88$ and $k = 3$, this produces an F statistic for significance of the independent variables of $F = [.1601/(1 - .1601)](84/3) \approx 5.34$. The associated p -value is .002, which is strong evidence against the null. The LM statistic is $88(.1601) \approx 14.09$; this gives a p -value $\approx .0028$ (using the χ^2_3 distribution), giving essentially the same conclusion as the F statistic. This means that the usual standard errors reported in (8.17) are not reliable.

In Chapter 6, we mentioned that one benefit of using the logarithmic functional form for the dependent variable is that heteroskedasticity is often reduced. In the current application, let us put *price*, *lotsize*, and *sqrft* in logarithmic form, so that the elasticities of *price*, with respect to *lotsize* and *sqrft*, are constant. The estimated equation is

$$\widehat{\log(\text{price})} = -1.30 + .168 \log(\text{lotsize}) + .700 \log(\text{sqrft}) + .037 \text{ bdrms} \\ (.65) \quad (.038) \quad (.093) \quad (.028) \quad [8.18] \\ n = 88, R^2 = .643.$$

Regressing the squared OLS residuals from this regression on $\log(\text{lotsize})$, $\log(\text{sqrft})$, and *bdrms* gives $R_u^2 = .0480$. Thus, $F = 1.41$ (p -value = .245), and $LM = 4.22$ (p -value = .239). Therefore, we fail to reject the null hypothesis of homoskedasticity in the model with the logarithmic functional forms. The occurrence of less heteroskedasticity with the dependent variable in logarithmic form has been noticed in many empirical applications.

If we suspect that heteroskedasticity depends only upon certain independent variables, we can easily modify the Breusch-Pagan test: we simply regress $\hat{u}^2$ on whatever independent variables we choose and carry out the appropriate F or LM test. Remember that the appropriate degrees of freedom depends upon the number of independent variables in the regression with $\hat{u}^2$ as the dependent variable; the number of independent variables showing up in equation (8.10) is irrelevant.

EXPLORING FURTHER 8.2

Consider wage equation (7.11), where you think that the conditional variance of $\log(\text{wage})$ does not depend on *educ*, *exper*, or *tenure*. However, you are worried that the variance of $\log(\text{wage})$ differs across the four demographic groups of married males, married females, single males, and single females. What regression would you run to test for heteroskedasticity? What are the degrees of freedom in the F test?

If the squared residuals are regressed on only a single independent variable, the test for heteroskedasticity is just the usual t statistic on the variable. A significant t statistic suggests that heteroskedasticity is a problem.

The White Test for Heteroskedasticity

In Chapter 5, we showed that the usual OLS standard errors and test statistics are asymptotically valid, provided all of the Gauss-Markov assumptions hold. It turns out that the homoskedasticity assumption, $\text{Var}(u_i|x_1, \dots, x_k) = \sigma^2$, can be replaced with the weaker assumption that the squared error, u^2 , is *uncorrelated* with all the independent variables (x_j), the squares of the independent variables (x_j^2), and all the cross products ($x_j x_h$ for $j \neq h$). This observation motivated White (1980) to propose a test for heteroskedasticity that adds the squares and cross products of all the independent variables to equation (8.14). The test is explicitly intended to test for forms of heteroskedasticity that invalidate the usual OLS standard errors and test statistics.

When the model contains $k = 3$ independent variables, the White test is based on an estimation of

$$\begin{aligned}\hat{u}^2 = & \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 \\ & + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + \text{error}.\end{aligned}\quad [8.19]$$

Compared with the Breusch-Pagan test, this equation has six more regressors. The **White test for heteroskedasticity** is the *LM* statistic for testing that all of the δ_j in equation (8.19) are zero, except for the intercept. Thus, nine restrictions are being tested in this case. We can also use an *F* test of this hypothesis; both tests have asymptotic justification.

With only three independent variables in the original model, equation (8.19) has nine independent variables. With six independent variables in the original model, the White regression would generally involve 27 regressors (unless some are redundant). This abundance of regressors is a weakness in the pure form of the White test: it uses many degrees of freedom for models with just a moderate number of independent variables.

It is possible to obtain a test that is easier to implement than the White test and more conserving on degrees of freedom. To create the test, recall that the difference between the White and Breusch-Pagan tests is that the former includes the squares and cross products of the independent variables. We can preserve the spirit of the White test while conserving on degrees of freedom by using the OLS fitted values in a test for heteroskedasticity. Remember that the fitted values are defined, for each observation i , by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_k x_{ik}.$$

These are just linear functions of the independent variables. If we square the fitted values, we get a particular function of all the squares and cross products of the independent variables. This suggests testing for heteroskedasticity by estimating the equation

$$\hat{u}^2 = \delta_0 + \delta_1 \hat{y} + \delta_2 \hat{y}^2 + \text{error},\quad [8.20]$$

where $\hat{y}$ stands for the fitted values. It is important not to confuse $\hat{y}$ and y in this equation. We use the fitted values because they are functions of the independent variables (and the estimated parameters); using y in (8.20) does not produce a valid test for heteroskedasticity.

We can use the *F* or *LM* statistic for the null hypothesis $H_0: \delta_1 = 0, \delta_2 = 0$ in equation (8.20). This results in two restrictions in testing the null of homoskedasticity, regardless of the number of independent variables in the original model. Conserving on degrees of freedom in this way is often a good idea, and it also makes the test easy to implement.

Since $\hat{y}$ is an estimate of the expected value of y , given the x_j , using (8.20) to test for heteroskedasticity is useful in cases where the variance is thought to change with the level of the expected value, $E(y|\mathbf{x})$. The test from (8.20) can be viewed as a special case of the White test, since equation (8.20) can be shown to impose restrictions on the parameters in equation (8.19).

A Special Case of the White Test for Heteroskedasticity:

1. Estimate the model (8.10) by OLS, as usual. Obtain the OLS residuals $\hat{u}$ and the fitted values $\hat{y}$. Compute the squared OLS residuals $\hat{u}^2$ and the squared fitted values $\hat{y}^2$.
2. Run the regression in equation (8.20). Keep the R -squared from this regression, R_a^2 .
3. Form either the F or LM statistic and compute the p -value (using the $F_{2,n-3}$ distribution in the former case and the χ^2_2 distribution in the latter case).

EXAMPLE 8.5

SPECIAL FORM OF THE WHITE TEST IN THE LOG HOUSING PRICE EQUATION

We apply the special case of the White test to equation (8.18), where we use the LM form of the statistic. The important thing to remember is that the chi-square distribution always has two df . The regression of $\hat{u}^2$ on $\ln \text{price}$, $(\ln \text{price})^2$, where $\ln \text{price}$ denotes the fitted values from (8.18), produces $R_a^2 = .0392$; thus, $LM = 88(.0392) \approx 3.45$, and the p -value = .178. This is stronger evidence of heteroskedasticity than is provided by the Breusch-Pagan test, but we still fail to reject homoskedasticity at even the 15% level.

Before leaving this section, we should discuss one important caveat. We have interpreted a rejection using one of the heteroskedasticity tests as evidence of heteroskedasticity. This is appropriate provided we maintain Assumptions MLR.1 through MLR.4. But, if MLR.4 is violated—in particular, if the functional form of $E(y|\mathbf{x})$ is misspecified—then a test for heteroskedasticity can reject H_0 , even if $\text{Var}(y|\mathbf{x})$ is constant. For example, if we omit one or more quadratic terms in a regression model or use the level model when we should use the log, a test for heteroskedasticity can be significant. This has led some economists to view tests for heteroskedasticity as general misspecification tests. However, there are better, more direct tests for functional form misspecification, and we will cover some of them in Section 9.1. It is better to use explicit tests for functional form first, since functional form misspecification is more important than heteroskedasticity. Then, once we are satisfied with the functional form, we can test for heteroskedasticity.

8.4 Weighted Least Squares Estimation

If heteroskedasticity is detected using one of the tests in Section 8.3, we know from Section 8.2 that one possible response is to use heteroskedasticity-robust statistics after estimation by OLS. Before the development of heteroskedasticity-robust statistics, the response to a finding of heteroskedasticity was to specify its form and use a *weighted least squares* method, which we develop in this section. As we will argue, if we have correctly specified the form of the variance (as a function of explanatory variables), then weighted least squares (WLS) is more efficient than OLS, and WLS leads to new t and F statistics

that have t and F distributions. We will also discuss the implications of using the wrong form of the variance in the WLS procedure.

The Heteroskedasticity Is Known up to a Multiplicative Constant

Let $\mathbf{x}$ denote all the explanatory variables in equation (8.10) and assume that

$$\text{Var}(u|\mathbf{x}) = \sigma^2 h(\mathbf{x}), \quad [8.21]$$

where $h(\mathbf{x})$ is some function of the explanatory variables that determines the heteroskedasticity. Since variances must be positive, $h(\mathbf{x}) > 0$ for all possible values of the independent variables. For now, we assume that the function $h(\mathbf{x})$ is known. The population parameter σ^2 is unknown, but we will be able to estimate it from a data sample.

For a random drawing from the population, we can write $\sigma_i^2 = \text{Var}(u_i|\mathbf{x}_i) = \sigma^2 h(\mathbf{x}_i) = \sigma^2 h_i$, where we again use the notation $\mathbf{x}_i$ to denote all independent variables for observation i , and h_i changes with each observation because the independent variables change across observations. For example, consider the simple savings function

$$\text{sav}_i = \beta_0 + \beta_1 \text{inc}_i + u_i \quad [8.22]$$

$$\text{Var}(u_i|\text{inc}_i) = \sigma^2 \text{inc}_i. \quad [8.23]$$

Here, $h(x) = h(\text{inc}) = \text{inc}$: the variance of the error is proportional to the level of income. This means that, as income increases, the variability in savings increases. (If $\beta_1 > 0$, the expected value of savings also increases with income.) Because inc is always positive, the variance in equation (8.23) is always guaranteed to be positive. The standard deviation of u_i , conditional on inc_i , is $\sigma\sqrt{\text{inc}_i}$.

How can we use the information in equation (8.21) to estimate the β_j ? Essentially, we take the original equation,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + u_i, \quad [8.24]$$

which contains heteroskedastic errors, and transform it into an equation that has homoskedastic errors (and satisfies the other Gauss-Markov assumptions). Since h_i is just a function of $\mathbf{x}_i$, $u_i/\sqrt{h_i}$ has a zero expected value conditional on $\mathbf{x}_i$. Further, since $\text{Var}(u_i|\mathbf{x}_i) = E(u_i^2|\mathbf{x}_i) = \sigma^2 h_i$, the variance of $u_i/\sqrt{h_i}$ (conditional on $\mathbf{x}_i$) is σ^2 :

$$E((u_i/\sqrt{h_i})^2) = E(u_i^2)/h_i = (\sigma^2 h_i)/h_i = \sigma^2,$$

where we have suppressed the conditioning on $\mathbf{x}_i$ for simplicity. We can divide equation (8.24) by $\sqrt{h_i}$ to get

$$\begin{aligned} y_i/\sqrt{h_i} &= \beta_0/\sqrt{h_i} + \beta_1(x_{i1}/\sqrt{h_i}) + \beta_2(x_{i2}/\sqrt{h_i}) + \dots \\ &\quad + \beta_k(x_{ik}/\sqrt{h_i}) + (u_i/\sqrt{h_i}) \end{aligned} \quad [8.25]$$

or

$$y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \dots + \beta_k x_{ik}^* + u_i^*, \quad [8.26]$$

where $x_{i0}^* = 1/\sqrt{h_i}$ and the other starred variables denote the corresponding original variables divided by $\sqrt{h_i}$.

Equation (8.26) looks a little peculiar, but the important thing to remember is that we derived it so we could obtain estimators of the β_j that have better efficiency properties than OLS. The intercept β_0 in the original equation (8.24) is now multiplying the variable $x_{i0}^* = 1/\sqrt{h_i}$. Each slope parameter in β_j multiplies a new variable that rarely has a useful interpretation. This should not cause problems if we recall that, for interpreting the parameters and the model, we always want to return to the original equation (8.24).

In the preceding savings example, the transformed equation looks like

$$sav_i/\sqrt{inc_i} = \beta_0(1/\sqrt{inc_i}) + \beta_1\sqrt{inc_i} + u_i^*,$$

where we use the fact that $inc_i/\sqrt{inc_i} = \sqrt{inc_i}$. Nevertheless, β_1 is the marginal propensity to save out of income, an interpretation we obtain from equation (8.22).

Equation (8.26) is linear in its parameters (so it satisfies MLR.1), and the random sampling assumption has not changed. Further, u_i^* has a zero mean and a constant variance (σ^2), conditional on $\mathbf{x}_i^*$. This means that if the original equation satisfies the first four Gauss-Markov assumptions, then the transformed equation (8.26) satisfies all five Gauss-Markov assumptions. Also, if u_i has a normal distribution, then u_i^* has a normal distribution with variance σ^2 . Therefore, the transformed equation satisfies the classical linear model assumptions (MLR.1 through MLR.6) if the original model does so except for the homoskedasticity assumption.

Since we know that OLS has appealing properties (is BLUE, for example) under the Gauss-Markov assumptions, the discussion in the previous paragraph suggests estimating the parameters in equation (8.26) by ordinary least squares. These estimators, β_0^* , β_1^* , ..., β_k^* , will be different from the OLS estimators in the original equation. The β_j^* are examples of **generalized least squares (GLS) estimators**. In this case, the GLS estimators are used to account for heteroskedasticity in the errors. We will encounter other GLS estimators in Chapter 12.

Because equation (8.26) satisfies all of the ideal assumptions, standard errors, t statistics, and F statistics can all be obtained from regressions using the transformed variables. The sum of squared residuals from (8.26) divided by the degrees of freedom is an unbiased estimator of σ^2 . Further, the GLS estimators, because they are the best linear unbiased estimators of the β_j , are necessarily more efficient than the OLS estimators $\hat{\beta}_j$ obtained from the untransformed equation. Essentially, after we have transformed the variables, we simply use standard OLS analysis. But we must remember to interpret the estimates in light of the original equation.

The R -squared that is obtained from estimating (8.26), while useful for computing F statistics, is not especially informative as a goodness-of-fit measure: it tells us how much variation in y^* is explained by the x_j^* , and this is seldom very meaningful.

The GLS estimators for correcting heteroskedasticity are called **weighted least squares (WLS) estimators**. This name comes from the fact that the β_j^* minimize the *weighted* sum of squared residuals, where each squared residual is weighted by $1/h_i$. The

idea is that less weight is given to observations with a higher error variance; OLS gives each observation the same weight because it is best when the error variance is identical for all partitions of the population. Mathematically, the WLS estimators are the values of the b_j that make

$$\sum_{i=1}^n (y_i - b_0 - b_1x_{i1} - b_2x_{i2} - \dots - b_kx_{ik})^2/h_i \quad [8.27]$$

as small as possible. Bringing the square root of $1/h_i$ inside the squared residual shows that the weighted sum of squared residuals is identical to the sum of squared residuals in the transformed variables:

$$\sum_{i=1}^n (y_i^* - b_0x_{i0}^* - b_1x_{i1}^* - b_2x_{i2}^* - \dots - b_kx_{ik}^*)^2.$$

Since OLS minimizes the sum of squared residuals (regardless of the definitions of the dependent variable and independent variable), it follows that the WLS estimators that minimize (8.27) are simply the OLS estimators from (8.26). Note carefully that the squared residuals in (8.27) are weighted by $1/h_i$, whereas the transformed variables in (8.26) are weighted by $1/\sqrt{h_i}$.

A weighted least squares estimator can be defined for any set of positive weights. OLS is the special case that gives equal weight to all observations. The efficient procedure, GLS, weights each squared residual by the *inverse* of the conditional variance of u_i given $\mathbf{x}_i$.

Obtaining the transformed variables in equation (8.25) in order to manually perform weighted least squares can be tedious, and the chance of making mistakes is nontrivial. Fortunately, most modern regression packages have a feature for computing weighted least squares. Typically, along with the dependent and independent variables in the original model, we just specify the weighting function, $1/h_i$, appearing in (8.27). That is, we specify weights proportional to the inverse of the variance. In addition to making mistakes less likely, this forces us to interpret weighted least squares estimates in the original model. In fact, we can write out the estimated equation in the usual way. The estimates and standard errors will be different from OLS, but the way we *interpret* those estimates, standard errors, and test statistics is the same.

EXAMPLE 8.6

FINANCIAL WEALTH EQUATION

We now estimate equations that explain net total financial wealth (*nettfa*, measured in \$1,000s) in terms of income (*inc*, also measured in \$1,000s) and some other variables, including age, gender, and an indicator for whether the person is eligible for a 401(k) pension plan. We use the data on single people (*fsize* = 1) in 401KSUBS.RAW. In Computer Exercise C12 in Chapter 6, it was found that a specific quadratic function in *age*, namely $(age - 25)^2$, fit the data just as well as an unrestricted quadratic. Plus, the restricted form gives a simplified interpretation because the minimum age in the sample is 25: *nettfa* is an increasing function of *age* after *age* = 25.

The results are reported in Table 8.1. Because we suspect heteroskedasticity, we report the heteroskedasticity-robust standard errors for OLS. The weighted least squares estimates, and their standard errors, are obtained under the assumption $\text{Var}(u|inc) = \sigma^2 inc$.

TABLE 8.1 Dependent Variable: *nettfa*

Independent Variables	(1) OLS	(2) WLS	(3) OLS	(4) WLS
<i>inc</i>	.821 (.104)	.787 (.063)	.771 (.100)	.740 (.064)
$(age - 25)^2$	—	—	.0251 (.0043)	.0175 (.0019)
<i>male</i>	—	—	2.48 (2.06)	1.84 (1.56)
<i>e401k</i>	—	—	6.89 (2.29)	5.19 (1.70)
<i>intercept</i>	-10.57 (2.53)	-9.58 (1.65)	-20.98 (3.50)	-16.70 (1.96)
Observations	2,017	2,017	2,017	2,017
R-squared	.0827	.0709	.1279	.1115

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Without controlling for other factors, another dollar of income is estimated to increase *nettfa* by about 82¢ when OLS is used; the WLS estimate is smaller, about 79¢. The difference is not large; we certainly do not expect them to be identical. The WLS coefficient does have a smaller standard error than OLS, almost 40% smaller, provided we assume the model $\text{Var}(\text{nettfa}|\text{inc}) = \sigma^2 \text{inc}$ is correct.

Adding the other controls reduced the *inc* coefficient somewhat, with the OLS estimate still larger than the WLS estimate. Again, the WLS estimate of β_{inc} is more precise. Age has an increasing effect starting at *age* = 25, with the OLS estimate showing a larger effect. The WLS estimate of β_{age} is more precise in this case. Gender does not have a statistically significant effect on *nettfa*, but being eligible for a 401(k) plan does: the OLS estimate is that those eligible, holding fixed income, age, and gender, have net total financial

assets about \$6,890 higher. The WLS estimate is substantially below the OLS estimate and suggests a misspecification of the functional form in the mean equation. (One possibility is to interact *e401k* and *inc*; see Computer Exercise C11.)

Using WLS, the *F* statistic for joint significance of $(age - 25)^2$, *male*, and *e401k* is about 30.8 if we use the

EXPLORING FURTHER 8.3

Using the OLS residuals obtained from the OLS regression reported in column (1) of Table 8.1, the regression of $\hat{u}^2$ on *inc* yields a *t* statistic of 2.96. Does it appear we should worry about heteroskedasticity in the financial wealth equation?

R-squareds reported in Table 8.1. With 2 and 2,012 degrees of freedom, the *p*-value is zero to more than 15 decimal places; of course, this is not surprising given the very large *t* statistics for the age and 401(k) variables.

Assuming that the error variance in the financial wealth equation has a variance proportional to income is essentially arbitrary. In fact, in most cases, our choice of weights in WLS has a degree of arbitrariness. However, there is one case where the weights needed for WLS arise naturally from an underlying econometric model. This happens when, instead of using

individual-level data, we only have averages of data across some group or geographic region. For example, suppose we are interested in determining the relationship between the amount a worker contributes to his or her 401(k) pension plan as a function of the plan generosity. Let i denote a particular firm and let e denote an employee within the firm. A simple model is

$$\text{contrib}_{i,e} = \beta_0 + \beta_1 \text{earn}_{i,e} + \beta_2 \text{age}_{i,e} + \beta_3 \text{mrte}_i + u_{i,e}, \quad [8.28]$$

where $\text{contrib}_{i,e}$ is the annual contribution by employee e who works for firm i , $\text{earn}_{i,e}$ is annual earnings for this person, and $\text{age}_{i,e}$ is the person's age. The variable mrte_i is the amount the firm puts into an employee's account for every dollar the employee contributes.

If (8.28) satisfies the Gauss-Markov assumptions, then we could estimate it, given a sample on individuals across various employers. Suppose, however, that we only have *average* values of contributions, earnings, and age by employer. In other words, individual-level data are not available. Thus, let $\overline{\text{contrib}}_i$ denote average contribution for people at firm i , and similarly for $\overline{\text{earn}}_i$ and $\overline{\text{age}}_i$. Let m_i denote the number of employees at firm i ; we assume that this is a known quantity. Then, if we average equation (8.28) across all employees at firm i , we obtain the firm-level equation

$$\overline{\text{contrib}}_i = \beta_0 + \beta_1 \overline{\text{earn}}_i + \beta_2 \overline{\text{age}}_i + \beta_3 \text{mrte}_i + \bar{u}_i, \quad [8.29]$$

where $\bar{u}_i = m_i^{-1} \sum_{e=1}^{m_i} u_{i,e}$ is the average error across all employees in firm i . If we have n firms in our sample, then (8.29) is just a standard multiple linear regression model that can be estimated by OLS. The estimators are unbiased if the original model (8.28) satisfies the Gauss-Markov assumptions and the individual errors $u_{i,e}$ are independent of the firm's size, m_i [because then the expected value of $\bar{u}_i$, given the explanatory variables in (8.29), is zero].

If the individual-level equation (8.28) satisfies the homoskedasticity assumption, and the errors within firm i are uncorrelated across employees, then we can show that the firm-level equation (8.29) has a particular kind of heteroskedasticity. Specifically, if $\text{Var}(u_{i,e}) = \sigma^2$ for all i and e , and $\text{Cov}(u_{i,e}, u_{i,g}) = 0$ for every pair of employees $e \neq g$ within firm i , then $\text{Var}(\bar{u}_i) = \sigma^2/m_i$; this is just the usual formula for the variance of an average of uncorrelated random variables with common variance. In other words, the variance of the error term $\bar{u}_i$ decreases with firm size. In this case, $h_i = 1/m_i$, and so the most efficient procedure is weighted least squares, with weights equal to the number of employees at the firm ($1/h_i = m_i$). This ensures that larger firms receive more weight. This gives us an efficient way of estimating the parameters in the individual-level model when we only have averages at the firm level.

A similar weighting arises when we are using per capita data at the city, county, state, or country level. If the individual-level equation satisfies the Gauss-Markov assumptions, then the error in the per capita equation has a variance proportional to one over the size of the population. Therefore, weighted least squares with weights equal to the population is appropriate. For example, suppose we have city-level data on per capita beer consumption (in ounces), the percentage of people in the population over 21 years old, average adult education levels, average income levels, and the city price of beer. Then, the city-level model

$$\text{beerpc} = \beta_0 + \beta_1 \text{perc21} + \beta_2 \text{avgeduc} + \beta_3 \text{incpc} + \beta_4 \text{price} + u$$

can be estimated by weighted least squares, with the weights being the city population.

The advantage of weighting by firm size, city population, and so on relies on the underlying individual equation being homoskedastic. If heteroskedasticity exists at the individual level, then the proper weighting depends on the form of heteroskedasticity. Further, if there is correlation across errors within a group (say, firm), then $\text{Var}(\bar{u}_i) \neq \sigma^2/m_i$; see Problem 7. Uncertainty about the form of $\text{Var}(\bar{u}_i)$ in equations such as (8.29) is why more and more researchers simply use OLS and compute robust standard errors and test statistics when estimating models using per capita data. An alternative is to weight by group size but to report the heteroskedasticity-robust statistics in the WLS estimation. This ensures that, while the estimation is efficient if the individual-level model satisfies the Gauss-Markov assumptions, heteroskedasticity at the individual level or within-group correlation are accounted for through robust inference.

The Heteroskedasticity Function Must Be Estimated: Feasible GLS

In the previous subsection, we saw some examples of where the heteroskedasticity is known up to a multiplicative form. In most cases, the exact form of heteroskedasticity is not obvious. In other words, it is difficult to find the function $h(\mathbf{x}_i)$ of the previous section. Nevertheless, in many cases we can model the function h and use the data to estimate the unknown parameters in this model. This results in an estimate of each h_i , denoted as $\hat{h}_i$. Using $\hat{h}_i$ instead of h_i in the GLS transformation yields an estimator called the **feasible GLS (FGLS) estimator**. Feasible GLS is sometimes called *estimated GLS*, or EGLS.

There are many ways to model heteroskedasticity, but we will study one particular, fairly flexible approach. Assume that

$$\text{Var}(u|\mathbf{x}) = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_k x_k), \quad [8.30]$$

where $x_1, x_2, \dots, x_k$ are the independent variables appearing in the regression model [see equation (8.1)], and the δ_j are unknown parameters. Other functions of the x_j can appear, but we will focus primarily on (8.30). In the notation of the previous subsection, $h(\mathbf{x}) = \exp(\delta_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_k x_k)$.

You may wonder why we have used the exponential function in (8.30). After all, when *testing* for heteroskedasticity using the Breusch-Pagan test, we assumed that heteroskedasticity was a linear function of the x_j . Linear alternatives such as (8.12) are fine when testing for heteroskedasticity, but they can be problematic when correcting for heteroskedasticity using weighted least squares. We have encountered the reason for this problem before: linear models do not ensure that predicted values are positive, and our estimated variances must be positive in order to perform WLS.

If the parameters δ_j were known, then we would just apply WLS, as in the previous subsection. This is not very realistic. It is better to use the data to estimate these parameters, and then to use these estimates to construct weights. How can we estimate the δ_j ? Essentially, we will transform this equation into a linear form that, with slight modification, can be estimated by OLS.

Under assumption (8.30), we can write

$$u^2 = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_k x_k) v,$$

where v has a mean equal to unity, conditional on $\mathbf{x} = (x_1, x_2, \dots, x_k)$. If we assume that v is actually independent of $\mathbf{x}$, we can write

$$\log(u^2) = a_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_k x_k + e, \quad [8.31]$$

where e has a zero mean and is independent of $\mathbf{x}$; the intercept in this equation is different from δ_0 , but this is not important in implementing WLS. The dependent variable is the log of the squared error. Since (8.31) satisfies the Gauss-Markov assumptions, we can get unbiased estimators of the δ_i by using OLS.

As usual, we must replace the unobserved u with the OLS residuals. Therefore, we run the regression of

$$\log(\hat{u}^2) \text{ on } x_1, x_2, \dots, x_k. \quad [8.32]$$

Actually, what we need from this regression are the fitted values; call these $\hat{g}_i$. Then, the estimates of h_i are simply

$$\hat{h}_i = \exp(\hat{g}_i). \quad [8.33]$$

We now use WLS with weights $1/\hat{h}_i$ in place of $1/h_i$ in equation (8.27). We summarize the steps.

A Feasible GLS Procedure to Correct for Heteroskedasticity:

1. Run the regression of y on $x_1, x_2, \dots, x_k$ and obtain the residuals, $\hat{u}$.
2. Create $\log(\hat{u}^2)$ by first squaring the OLS residuals and then taking the natural log.
3. Run the regression in equation (8.32) and obtain the fitted values, $\hat{g}$.
4. Exponentiate the fitted values from (8.32): $\hat{h} = \exp(\hat{g})$.
5. Estimate the equation

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$$

by WLS, using weights $1/\hat{h}$. In other words, we replace h_i with $\hat{h}_i$ in equation (8.27). Remember, the *squared* residual for observation i gets weighted by $1/\hat{h}_i$. If instead we first transform all variables and run OLS, each variable gets multiplied by $1/\sqrt{\hat{h}_i}$, including the intercept.

If we could use h_i rather than $\hat{h}_i$ in the WLS procedure, we know that our estimators would be unbiased; in fact, they would be the best linear unbiased estimators, assuming that we have properly modeled the heteroskedasticity. Having to estimate h_i using the same data means that the FGLS estimator is no longer unbiased (so it cannot be BLUE, either). Nevertheless, the FGLS estimator is consistent and *asymptotically* more efficient than OLS. This is difficult to show because of estimation of the variance parameters. But if we ignore this—as it turns out we may—the proof is similar to showing that OLS is efficient in the class of estimators in Theorem 5.3. At any rate, for large sample sizes, FGLS is an attractive alternative to OLS when there is evidence of heteroskedasticity that inflates the standard errors of the OLS estimates.

We must remember that the FGLS estimators are estimators of the parameters in the usual population model

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u.$$

Just as the OLS estimates measure the marginal impact of each x_j on y , so do the FGLS estimates. We use the FGLS estimates in place of the OLS estimates because the FGLS estimators are more efficient and have associated test statistics with the usual t and F distributions, at least in large samples. If we have some doubt about the variance specified in equation (8.30), we can use heteroskedasticity-robust standard errors and test statistics in the transformed equation.

Another useful alternative for estimating h_i is to replace the independent variables in regression (8.32) with the OLS fitted values and their squares. In other words, obtain the $\hat{g}_i$ as the fitted values from the regression of

$$\log(\hat{u}^2) \text{ on } \hat{y}, \hat{y}^2 \quad [8.34]$$

and then obtain the $\hat{h}_i$ exactly as in equation (8.33). This changes only step (3) in the previous procedure.

If we use regression (8.32) to estimate the variance function, you may be wondering if we can simply test for heteroskedasticity using this same regression (an F or LM test can be used). In fact, Park (1966) suggested this. Unfortunately, when compared with the tests discussed in Section 8.3, the Park test has some problems. First, the null hypothesis must be something stronger than homoskedasticity: effectively, u and $\mathbf{x}$ must be independent. This is not required in the Breusch-Pagan or White tests. Second, using the OLS residuals $\hat{u}$ in place of u in (8.32) can cause the F statistic to deviate from the F distribution, even in large sample sizes. This is not an issue in the other tests we have covered. For these reasons, the Park test is not recommended when testing for heteroskedasticity. Regression (8.32) works well for weighted least squares because we only need consistent estimators of the δ_j , and regression (8.32) certainly delivers those.

EXAMPLE 8.7

DEMAND FOR CIGARETTES

We use the data in SMOKE.RAW to estimate a demand function for daily cigarette consumption. Since most people do not smoke, the dependent variable, *cigs*, is zero for most observations. A linear model is not ideal because it can result in negative predicted values. Nevertheless, we can still learn something about the determinants of cigarette smoking by using a linear model.

The equation estimated by ordinary least squares, with the usual OLS standard errors in parentheses, is

$$\begin{aligned} \widehat{cigs} = & -3.64 + .880 \log(income) - .751 \log(cigpric) \\ & (24.08) \quad (.728) \quad (5.773) \\ & - .501 educ + .771 age - .0090 age^2 - 2.83 restaurn \\ & (.167) \quad (.160) \quad (.0017) \quad (1.11) \\ n = & 807, R^2 = .0526, \end{aligned} \quad [8.35]$$

where

- cigs* = number of cigarettes smoked per day.
- income* = annual income.
- cigpric* = the per-pack price of cigarettes (in cents).
- educ* = years of schooling.
- age* = measured in years.

restaurn = a binary indicator equal to unity if the person resides in a state with restaurant smoking restrictions.

Since we are also going to do weighted least squares, we do not report the heteroskedasticity-robust standard errors for OLS. (Incidentally, 13 out of the 807 fitted values are less than zero; this is less than 2% of the sample and is not a major cause for concern.)

Neither income nor cigarette price is statistically significant in (8.35), and their effects are not practically large. For example, if income increases by 10%, *cigs* is predicted to increase by $(.880/100)(10) = .088$, or less than one-tenth of a cigarette per day. The magnitude of the price effect is similar.

Each year of education reduces the average cigarettes smoked per day by one-half of a cigarette, and the effect is statistically significant. Cigarette smoking is also related to age, in a quadratic fashion. Smoking increases with age up until $age = .771/[2(.009)] \approx 42.83$, and then smoking decreases with age. Both terms in the quadratic are statistically significant. The presence of a restriction on smoking in restaurants decreases cigarette smoking by almost three cigarettes per day, on average.

Do the errors underlying equation (8.35) contain heteroskedasticity? The Breusch-Pagan regression of the squared OLS residuals on the independent variables in (8.35) [see equation (8.14)] produces $R_a^2 = .040$. This small R -squared may seem to indicate no heteroskedasticity, but we must remember to compute either the F or LM statistic. If the sample size is large, a seemingly small R_a^2 can result in a very strong rejection of homoskedasticity. The LM statistic is $LM = 807(.040) = 32.28$, and this is the outcome of a χ^2_6 random variable. The p -value is less than .000015, which is very strong evidence of heteroskedasticity.

Therefore, we estimate the equation using the feasible GLS procedure based on equation (8.32). The weighted least squares estimates are

$$\begin{aligned}\widehat{cigs} = & 5.64 + 1.30 \log(\text{income}) - 2.94 \log(\text{cigpric}) \\ & (17.80) \quad (.44) \quad (4.46) \\ & - .463 \text{educ} + .482 \text{age} - .0056 \text{age}^2 - 3.46 \text{restaurn} \\ & (.120) \quad (.097) \quad (.0009) \quad (.80) \\ n = & 807, R^2 = .1134.\end{aligned}\tag{8.36}$$

The income effect is now statistically significant and larger in magnitude. The price effect is also notably bigger, but it is still statistically insignificant. [One reason for this is that *cigpric* varies only across states in the sample, and so there is much less variation in $\log(\text{cigpric})$ than in $\log(\text{income})$, *educ*, and *age*.]

The estimates on the other variables have, naturally, changed somewhat, but the basic story is still the same. Cigarette smoking is negatively related to schooling, has a quadratic relationship with *age*, and is negatively affected by restaurant smoking restrictions.

We must be a little careful in computing F statistics for testing multiple hypotheses after estimation by WLS. (This is true whether the sum of squared residuals or R -squared form of the F statistic is used.) It is important that the same weights be used to estimate the unrestricted and restricted models. We should first estimate the unrestricted model by OLS. Once we have obtained the weights, we can use them to estimate the restricted

model as well. The F statistic can be computed as usual. Fortunately, many regression packages have a simple command for testing joint restrictions after WLS estimation, so we need not perform the restricted regression ourselves.

Example 8.7 hints at an issue that sometimes arises in applications of weighted least squares: the OLS and WLS estimates can be substantially different. This is not such a big problem in the demand for cigarettes equation because all the coefficients maintain the

same signs, and the biggest changes are on variables that were statistically insignificant when the equation was estimated by OLS. The OLS and WLS estimates will always differ due to sampling error. The issue is whether their difference is enough to change important conclusions.

If OLS and WLS produce statistically significant estimates that differ in sign—for example, the OLS price elasticity is positive and significant, while the WLS price elasticity is negative and significant—or the difference in magnitudes of the estimates is practically large, we should be suspicious. Typically, this indicates that one of the *other* Gauss-Markov assumptions is false, particularly the zero conditional mean assumption on the error (MLR.4). If $E(y|\mathbf{x}) \neq \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$, then OLS and WLS have different expected values and probability limits. For WLS to be con-

EXPLORING FURTHER 8.4

Let $\hat{u}_i$ be the WLS residuals from (8.36), which are not weighted, and let $\widehat{cigs}_i$ be the fitted values. (These are obtained using the same formulas as OLS; they differ because of different estimates of the β_j .) One way to determine whether heteroskedasticity has been eliminated is to use the $\hat{u}_i^2/\hat{h}_i = (\hat{u}_i/\sqrt{\hat{h}_i})^2$ in a test for heteroskedasticity. [If $h_i = \text{Var}(u_i|\mathbf{x}_i)$, then the transformed residuals should have little evidence of heteroskedasticity.] There are many possibilities, but one—based on White’s test in the transformed equation—is to regress $\hat{u}_i^2/\hat{h}_i$ on $\widehat{cigs}_i/\sqrt{\hat{h}_i}$ and $\widehat{cigs}_i^2/\hat{h}_i$ (including an intercept). The joint F statistic when we use SMOKE.RAW is 11.15. Does it appear that our correction for heteroskedasticity has actually eliminated the heteroskedasticity?

sistent for the β_j , it is not enough for u to be uncorrelated with each x_j ; we need the stronger assumption MLR.4 in the linear model MLR.1. Therefore, a significant difference between OLS and WLS can indicate a functional form misspecification in $E(y|\mathbf{x})$. The *Hausman test* [Hausman (1978)] can be used to formally compare the OLS and WLS estimates to see if they differ by more than sampling error suggests they should, but this test is beyond the scope of this text. In many cases, an informal “eyeballing” of the estimates is sufficient to detect a problem.

What If the Assumed Heteroskedasticity Function Is Wrong?

We just noted that if OLS and WLS produce very different estimates, it is likely that the conditional mean $E(y|\mathbf{x})$ is misspecified. What are the properties of WLS if the variance function we use is misspecified in the sense that $\text{Var}(y|\mathbf{x}) \neq \sigma^2 h(\mathbf{x})$ for our chosen function $h(\mathbf{x})$? The most important issue is whether misspecification of $h(\mathbf{x})$ causes bias or inconsistency in the WLS estimator. Fortunately, the answer is no, at least under MLR.4. Recall that, if $E(u|\mathbf{x}) = 0$, then any function of $\mathbf{x}$ is uncorrelated with u , and so the weighted error, $u/\sqrt{h(\mathbf{x})}$, is uncorrelated with the weighted regressors, $x_j/\sqrt{h(\mathbf{x})}$, for *any* function $h(\mathbf{x})$ that is always positive. This is why, as we just discussed, we can take large differences between the OLS and WLS estimators as indicative of functional form misspecification. If we estimate parameters in the function, say $h(\mathbf{x}, \hat{\delta})$, then we can no longer claim that WLS is unbiased, but it will generally be consistent (whether or not the variance function is correctly specified).

If WLS is at least consistent under MLR.1 to MLR.4, what are the consequences of using WLS with a misspecified variance function? There are two. The first, which is very important, is that the usual WLS standard errors and test statistics, computed under the assumption that $\text{Var}(y|\mathbf{x}) = \sigma^2 h(\mathbf{x})$, are no longer valid, even in large samples. For example, the WLS estimates and standard errors in column (4) of Table 8.1 assume that $\text{Var}(\text{netffa}|\text{inc}, \text{age}, \text{male}, \text{e401k}) = \text{Var}(\text{netffa}|\text{inc}) = \sigma^2 \text{inc}$; so we are assuming not only that the variance depends just on income, but also that it is a linear function of income. If this assumption is false, the standard errors (and any statistics we obtain using those standard errors) are not valid. Fortunately, there is an easy fix: just as we can obtain standard errors for the OLS estimates that are robust to arbitrary heteroskedasticity, we can obtain standard errors for WLS that allow the variance function to be arbitrarily misspecified. It is easy to see why this works. Write the transformed equation as

$$y_i/\sqrt{h_i} = \beta_0(1/\sqrt{h_i}) + \beta_1(x_{i1}/\sqrt{h_i}) + \dots + \beta_k(x_{ik}/\sqrt{h_i}) + u_i/\sqrt{h_i}.$$

Now, if $\text{Var}(u_i|\mathbf{x}_i) \neq \sigma^2 h_i$, then the weighted error $u_i/\sqrt{h_i}$ is heteroskedastic. So we can just apply the usual heteroskedasticity-robust standard errors after estimating this equation by OLS—which, remember, is identical to WLS.

To see how robust inference with WLS works in practice, column (1) of Table 8.2 reproduces the last column of Table 8.1, and column (2) contains standard errors robust to $\text{Var}(u_i|\mathbf{x}_i) \neq \sigma^2 \text{inc}_i$.

The standard errors in column (2) allow the variance function to be misspecified. We see that, for the income and age variables, the robust standard errors are somewhat above the usual WLS standard errors—certainly by enough to stretch the confidence intervals. On the other hand, the robust standard errors for *male* and *e401k* are actually smaller than those that assume a correct variance function. We saw this could happen with the heteroskedasticity-robust standard errors for OLS, too.

Even if we use flexible forms of variance functions, such as that in (8.30), there is no guarantee that we have the correct model. While exponential heteroskedasticity is

TABLE 8.2 WLS Estimation of the *netffa* Equation

Independent Variables	With Nonrobust Standard Errors	With Robust Standard Errors
<i>inc</i>	.740 (.064)	.740 (.075)
$(\text{age} - 25)^2$	.0175 (.0019)	.0175 (.0026)
<i>male</i>	1.84 (1.56)	1.84 (1.31)
<i>e401k</i>	5.19 (1.70)	5.19 (1.57)
<i>intercept</i>	-16.70 (1.96)	-16.70 (2.24)
Observations	2,017	2,017
<i>R</i> -squared	.1115	.1115

appealing and reasonably flexible, it is, after all, just a model. Therefore, it is always a good idea to compute fully robust standard errors and test statistics after WLS estimation.

A modern criticism of WLS is that if the variance function is misspecified, it is not guaranteed to be more efficient than OLS. In fact, that is the case: if $\text{Var}(y|\mathbf{x})$ is neither constant nor equal to $\sigma^2 h(\mathbf{x})$, where $h(\mathbf{x})$ is the proposed model of heteroskedasticity, then we cannot rank OLS and WLS in terms of variances (or asymptotic variances when the variance parameters must be estimated). However, this theoretically correct criticism misses an important practical point. Namely, in cases of strong heteroskedasticity, it is often better to use a wrong form of heteroskedasticity and apply WLS than to ignore heteroskedasticity altogether in estimation and use OLS. Models such as (8.30) can well approximate a variety of heteroskedasticity functions and may produce estimators with smaller (asymptotic) variances. Even in Example 8.6, where the form of heteroskedasticity was assumed to have the simple form $\text{Var}(\text{netffa}|\mathbf{x}) = \sigma^2 \text{inc}$, the fully robust standard errors for WLS are well below the fully robust standard errors for OLS. (Comparing robust standard errors for the two estimators puts them on equal footing: we assume neither homoskedasticity nor that the variance has the form $\sigma^2 \text{inc}$.) For example, the robust standard error for the WLS estimator of β_{inc} is about .075, which is 25% lower than the robust standard error for OLS (about .100). For the coefficient on $(\text{age} - 25)^2$, the robust standard error of WLS is about .0026, almost 40% below the robust standard error for OLS (about .0043).

Prediction and Prediction Intervals with Heteroskedasticity

If we start with the standard linear model under MLR.1 to MLR.4, but allow for heteroskedasticity of the form $\text{Var}(y|\mathbf{x}) = \sigma^2 h(\mathbf{x})$ [see equation (8.21)], the presence of heteroskedasticity affects the point prediction of y only insofar as it affects estimation of the β_j . Of course, it is natural to use WLS on a sample of size n to obtain the $\hat{\beta}_j$. Our prediction of an unobserved outcome, y^0 , given known values of the explanatory variables $\mathbf{x}^0$, has the same form as in Section 6.4: $\hat{y}^0 = \hat{\beta}_0 + \mathbf{x}^0 \hat{\boldsymbol{\beta}}$. This makes sense: once we know $E(y|\mathbf{x})$, we base our prediction on it; the structure of $\text{Var}(y|\mathbf{x})$ plays no direct role.

On the other hand, prediction *intervals* do depend directly on the nature of $\text{Var}(y|\mathbf{x})$. Recall in Section 6.4 that we constructed a prediction interval under the classical linear model assumptions. Suppose now that all the CLM assumptions hold except that (8.21) replaces the homoskedasticity assumption, MLR.5. We know that the WLS estimators are BLUE and, because of normality, have (conditional) normal distributions. We can obtain $\text{se}(\hat{y}^0)$ using the same method in Section 6.4, except that now we use WLS. [A simple approach is to write $y_i = \theta_0 + \beta_1(x_{i1} - x_1^0) + \dots + \beta_k(x_{ik} - x_k^0) + u_i$, where the x_j^0 are the values of the explanatory variables for which we want a predicted value of y . We can estimate this equation by WLS and then obtain $\hat{y}^0 = \hat{\theta}_0$ and $\text{se}(\hat{y}^0) = \text{se}(\hat{\theta}_0)$.] We also need to estimate the standard deviation of u^0 , the unobserved part of y^0 . But $\text{Var}(u^0|\mathbf{x} = \mathbf{x}^0) = \sigma^2 h(\mathbf{x}^0)$, and so $\text{se}(u^0) = \hat{\sigma} \sqrt{h(\mathbf{x}^0)}$, where $\hat{\sigma}$ is the standard error of the regression from the WLS estimation. Therefore, a 95% prediction interval is

$$\hat{y}^0 \pm t_{.025} \cdot \text{se}(\hat{e}^0) \quad [8.37]$$

where $\text{se}(\hat{e}^0) = \{[\text{se}(\hat{y}^0)]^2 + \hat{\sigma}^2 h(\mathbf{x}^0)\}^{1/2}$.

This interval is exact only if we do not have to estimate the variance function. If we estimate parameters, as in model (8.30), then we cannot obtain an exact interval. In fact,

accounting for the estimation error in the $\hat{\beta}_j$ and the $\hat{\delta}_j$ (the variance parameters) becomes very difficult. We saw two examples in Section 6.4 where the estimation error in the parameters was swamped by the variation in the unobservables, u^0 . Therefore, we might still use equation (8.37) with $h(\mathbf{x}^0)$ simply replaced by $\hat{h}(\mathbf{x}^0)$. In fact, if we are to ignore the parameter estimation error entirely, we can drop $\text{se}(\hat{y}^0)$ from $\text{se}(\hat{e}^0)$. [Remember, $\text{se}(\hat{y}^0)$ converges to zero at the rate $1/\sqrt{n}$, while $\text{se}(\hat{u}^0)$ is roughly constant.]

We can also obtain a prediction for y in the model

$$\log(y) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u, \quad [8.38]$$

where u is heteroskedastic. We assume that u has a conditional normal distribution with a specific form of heteroskedasticity. We assume the exponential form in equation (8.30), but add the normality assumption:

$$u|x_1, x_2, \dots, x_k \sim \text{Normal}[0, \exp(\delta_0 + \delta_1 x_1 + \dots + \delta_k x_k)]. \quad [8.39]$$

As a notational shorthand, write the variance function as $\exp(\delta_0 + \mathbf{x}\boldsymbol{\delta})$. Then, because $\log(y)$ given $\mathbf{x}$ has a normal distribution with mean $\beta_0 + \mathbf{x}\boldsymbol{\beta}$ and variance $\exp(\delta_0 + \mathbf{x}\boldsymbol{\delta})$, it follows that

$$E(y|\mathbf{x}) = \exp(\beta_0 + \mathbf{x}\boldsymbol{\beta} + \sigma^2 \exp(\delta_0 + \mathbf{x}\boldsymbol{\delta})/2). \quad [8.40]$$

Now we estimate the β_j and δ_j using WLS estimation of (8.38). That is, after OLS to obtain the residuals, run the regression in (8.32) to obtain fitted values,

$$\hat{g}_i = \hat{\alpha}_0 + \hat{\delta}_1 x_{i1} + \dots + \hat{\delta}_k x_{ik}, \quad [8.41]$$

and then the $\hat{h}_i$ as in (8.33). Using these $\hat{h}_i$, obtain the WLS estimates, $\hat{\beta}_j$ and also $\hat{\sigma}^2$. Then, for each i , we can obtain a fitted value

$$\hat{y}_i = \exp(\widehat{\log y}_i + \hat{\sigma}^2 \hat{h}_i/2). \quad [8.42]$$

We can use these fitted values to obtain an R -squared measure, as described in Section 6.4: use the squared correlation coefficient between y_i and $\hat{y}_i$.

For any values of the explanatory variables $\mathbf{x}^0$, we can estimate $E(y|\mathbf{x} = \mathbf{x}^0)$ as

$$\hat{E}(y|\mathbf{x} = \mathbf{x}^0) = \exp(\hat{\beta}_0 + \mathbf{x}^0 \hat{\boldsymbol{\beta}} + \hat{\sigma}^2 \exp(\hat{\alpha}_0 + \mathbf{x}^0 \hat{\boldsymbol{\delta}})/2), \quad [8.43]$$

where

- $\hat{\beta}_j$ = the WLS estimates.
- $\hat{\alpha}_0$ = the intercept in (8.41).
- $\hat{\delta}_j$ = the slopes from the same regression.

Obtaining a proper standard error for the prediction in (8.42) is very complicated analytically, but, as in Section 6.4, it would be fairly easy to obtain a standard error using a resampling method such as the bootstrap described in Appendix 6A.

Obtaining a prediction interval is more of a challenge when we estimate a model for heteroskedasticity, and a full treatment is complicated. Nevertheless, we saw in Section 6.4 two examples where the error variance swamps the estimation error, and we would make only a small mistake by ignoring the estimation error in all parameters. Using arguments similar to those in Section 6.4, an approximate 95% prediction interval (for large sample

sizes) is $\exp[-1.96 \cdot \hat{\sigma} \sqrt{\hat{h}(\mathbf{x}^0)}] \exp(\hat{\beta}_0 + \mathbf{x}^0 \hat{\boldsymbol{\beta}})$ to $\exp[1.96 \cdot \hat{\sigma} \sqrt{\hat{h}(\mathbf{x}^0)}] \exp(\hat{\beta}_0 + \mathbf{x}^0 \hat{\boldsymbol{\beta}})$, where $\hat{h}(\mathbf{x}^0)$ is the estimated variance function evaluated at $\mathbf{x}^0$, $\hat{h}(\mathbf{x}^0) = \exp(\hat{\alpha}_0 + \delta_1 x_1^0 + \dots + \delta_k x_k^0)$. As in Section 6.4, we obtain this approximate interval by simply exponentiating the endpoints.

8.5 The Linear Probability Model Revisited

As we saw in Section 7.5, when the dependent variable y is a binary variable, the model must contain heteroskedasticity, unless all of the slope parameters are zero. We are now in a position to deal with this problem.

The simplest way to deal with heteroskedasticity in the linear probability model is to continue to use OLS estimation, but to also compute robust standard errors in test statistics. This ignores the fact that we actually know the form of heteroskedasticity for the LPM. Nevertheless, OLS estimation of the LPM is simple and often produces satisfactory results.

EXAMPLE 8.8

LABOR FORCE PARTICIPATION OF MARRIED WOMEN

In the labor force participation example in Section 7.5 [see equation (7.29)], we reported the usual OLS standard errors. Now, we compute the heteroskedasticity-robust standard errors as well. These are reported in brackets below the usual standard errors:

$$\begin{aligned} \widehat{inlf} = & .586 - .0034 \text{nwifeinc} + .038 \text{educ} + .039 \text{exper} \\ & (.154) \quad (.0014) \quad (.007) \quad (.006) \\ & [.151] \quad [.0015] \quad [.007] \quad [.006] \\ & - .00060 \text{exper}^2 - .016 \text{age} - .262 \text{kidslt6} + .0130 \text{kidsge6} \quad [8.44] \\ & (.00018) \quad (.002) \quad (.034) \quad (.0132) \\ & [.00019] \quad [.002] \quad [.032] \quad [.0135] \\ n = & 753, R^2 = .264. \end{aligned}$$

Several of the robust and OLS standard errors are the same to the reported degree of precision; in all cases, the differences are practically very small. Therefore, while heteroskedasticity is a problem in theory, it is not in practice, at least not for this example. It often turns out that the usual OLS standard errors and test statistics are similar to their heteroskedasticity-robust counterparts. Furthermore, it requires a minimal effort to compute both.

Generally, the OLS estimators are inefficient in the LPM. Recall that the conditional variance of y in the LPM is

$$\text{Var}(y|\mathbf{x}) = p(\mathbf{x})[1 - p(\mathbf{x})], \quad [8.45]$$

where

$$p(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k \quad [8.46]$$

is the response probability (probability of success, $y = 1$). It seems natural to use weighted least squares, but there are a couple of hitches. The probability $p(\mathbf{x})$ clearly depends on the unknown population parameters, β_j . Nevertheless, we do have unbiased estimators of these parameters, namely the OLS estimators. When the OLS estimators are plugged into equation (8.46), we obtain the OLS fitted values. Thus, for each observation i , $\text{Var}(y_i|\mathbf{x}_i)$ is estimated by

$$\hat{h}_i = \hat{y}_i(1 - \hat{y}_i), \quad [8.47]$$

where $\hat{y}_i$ is the OLS fitted value for observation i . Now, we apply feasible GLS, just as in Section 8.4.

Unfortunately, being able to estimate h_i for each i does not mean that we can proceed directly with WLS estimation. The problem is one that we briefly discussed in Section 7.5: the fitted values $\hat{y}_i$ need not fall in the unit interval. If either $\hat{y}_i < 0$ or $\hat{y}_i > 1$, equation (8.47) shows that $\hat{h}_i$ will be negative. Since WLS proceeds by multiplying observation i by $1/\sqrt{\hat{h}_i}$, the method will fail if $\hat{h}_i$ is negative (or zero) for any observation. In other words, all of the weights for WLS must be positive.

In some cases, $0 < \hat{y}_i < 1$ for all i , in which case WLS can be used to estimate the LPM. In cases with many observations and small probabilities of success or failure, it is very common to find some fitted values outside the unit interval. If this happens, as it does in the labor force participation example in equation (8.44), it is easiest to abandon WLS and to report the heteroskedasticity-robust statistics. An alternative is to adjust those fitted values that are less than zero or greater than unity, and then to apply WLS. One suggestion is to set $\hat{y}_i = .01$ if $\hat{y}_i < 0$ and $\hat{y}_i = .99$ if $\hat{y}_i > 1$. Unfortunately, this requires an arbitrary choice on the part of the researcher—for example, why not use .001 and .999 as the adjusted values? If many fitted values are outside the unit interval, the adjustment to the fitted values can affect the results; in this situation, it is probably best to just use OLS.

Estimating the Linear Probability Model by Weighted Least Squares:

1. Estimate the model by OLS and obtain the fitted values, $\hat{y}$.
2. Determine whether all of the fitted values are inside the unit interval. If so, proceed to step (3). If not, some adjustment is needed to bring all fitted values into the unit interval.
3. Construct the estimated variances in equation (8.47).
4. Estimate the equation

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$$

by WLS, using weights $1/\hat{h}$.

EXAMPLE 8.9

DETERMINANTS OF PERSONAL COMPUTER OWNERSHIP

We use the data in GPA1.RAW to estimate the probability of owning a computer. Let PC denote a binary indicator equal to unity if the student owns a computer, and zero otherwise. The variable $hsGPA$ is high school GPA, ACT is achievement test score, and $parcoll$ is a binary indicator equal to unity if at least one parent attended college. (Separate college indicators for the mother and the father do not yield individually significant results, as these are pretty highly correlated.)

The equation estimated by OLS is

$$\begin{aligned}\widehat{PC} &= -.0004 + .065 \text{ hsGPA} + .0006 \text{ ACT} + .221 \text{ parcoll} \\ &\quad (.4905) \quad (.137) \quad (.0155) \quad (.093) \\ &\quad [.4888] \quad [.139] \quad [.0158] \quad [.087] \\ n &= 141, R^2 = .0415.\end{aligned}\tag{8.48}$$

Just as with Example 8.8, there are no striking differences between the usual and robust standard errors. Nevertheless, we also estimate the model by WLS. Because all of the OLS fitted values are inside the unit interval, no adjustments are needed:

$$\begin{aligned}\widehat{PC} &= .026 + .033 \text{ hsGPA} + .0043 \text{ ACT} + .215 \text{ parcoll} \\ &\quad (.477) \quad (.130) \quad (.0155) \quad (.086) \\ n &= 141, R^2 = .0464.\end{aligned}\tag{8.49}$$

There are no important differences in the OLS and WLS estimates. The only significant explanatory variable is *parcoll*, and in both cases we estimate that the probability of PC ownership is about .22 higher if at least one parent attended college.

Summary

We began by reviewing the properties of ordinary least squares in the presence of heteroskedasticity. Heteroskedasticity does not cause bias or inconsistency in the OLS estimators, but the usual standard errors and test statistics are no longer valid. We showed how to compute heteroskedasticity-robust standard errors and *t* statistics, something that is routinely done by many regression packages. Most regression packages also compute a heteroskedasticity-robust *F*-type statistic.

We discussed two common ways to test for heteroskedasticity: the Breusch-Pagan test and a special case of the White test. Both of these statistics involve regressing the *squared* OLS residuals on either the independent variables (BP) or the fitted and squared fitted values (White). A simple *F* test is asymptotically valid; there are also Lagrange multiplier versions of the tests.

OLS is no longer the best linear unbiased estimator in the presence of heteroskedasticity. When the form of heteroskedasticity is known, generalized least squares (GLS) estimation can be used. This leads to weighted least squares as a means of obtaining the BLUE estimator. The test statistics from the WLS estimation are either exactly valid when the error term is normally distributed or asymptotically valid under nonnormality. This assumes, of course, that we have the proper model of heteroskedasticity.

More commonly, we must estimate a model for the heteroskedasticity before applying WLS. The resulting *feasible* GLS estimator is no longer unbiased, but it is consistent and asymptotically efficient. The usual statistics from the WLS regression are asymptotically valid. We discussed a method to ensure that the estimated variances are strictly positive for all observations, something needed to apply WLS.

As we discussed in Chapter 7, the linear probability model for a binary dependent variable necessarily has a heteroskedastic error term. A simple way to deal with this problem is to compute heteroskedasticity-robust statistics. Alternatively, if all the fitted values (that is, the estimated probabilities) are strictly between zero and one, weighted least squares can be used to obtain asymptotically efficient estimators.

Key Terms

Breusch-Pagan Test for Heteroskedasticity (BP Test)	Heteroskedasticity-Robust F Statistic	Weighted Least Squares (WLS) Estimators
Feasible GLS (FGLS) Estimator	Heteroskedasticity-Robust LM Statistic	White Test for Heteroskedasticity
Generalized Least Squares (GLS) Estimators	Heteroskedasticity-Robust Standard Error	
Heteroskedasticity of Unknown Form	Heteroskedasticity-Robust t Statistic	

Problems

- Which of the following are consequences of heteroskedasticity?
 - The OLS estimators, $\hat{\beta}_j$, are inconsistent.
 - The usual F statistic no longer has an F distribution.
 - The OLS estimators are no longer BLUE.

- Consider a linear model to explain monthly beer consumption:

$$beer = \beta_0 + \beta_1 inc + \beta_2 price + \beta_3 educ + \beta_4 female + u$$

$$E(u|inc, price, educ, female) = 0$$

$$\text{Var}(u|inc, price, educ, female) = \sigma^2 inc^2.$$

Write the transformed equation that has a homoskedastic error term.

- True or False: WLS is preferred to OLS when an important variable has been omitted from the model.
- Using the data in GPA3.RAW, the following equation was estimated for the fall and second semester students:

$$\begin{aligned} \widehat{trmgpa} = & -2.12 + .900 crsgpa + .193 cumgpa + .0014 tothrs \\ & (.55) (.175) \quad (.064) \quad (.0012) \\ & [.55] [.166] \quad [.074] \quad [.0012] \\ & + .0018 sat - .0039 hspcr + .351 female - .157 season \\ & (.0002) \quad (.0018) \quad (.085) \quad (.098) \\ & [.0002] \quad [.0019] \quad [.079] \quad [.080] \\ n = & 269, R^2 = .465. \end{aligned}$$

Here, *trmgpa* is term GPA, *crsgpa* is a weighted average of overall GPA in courses taken, *cumgpa* is GPA prior to the current semester, *tothrs* is total credit hours prior to the semester, *sat* is SAT score, *hsperc* is graduating percentile in high school class, *female* is a gender dummy, and *season* is a dummy variable equal to unity if the student's sport is in season during the fall. The usual and heteroskedasticity-robust standard errors are reported in parentheses and brackets, respectively.

- (i) Do the variables *crsgpa*, *cumgpa*, and *tothrs* have the expected estimated effects? Which of these variables are statistically significant at the 5% level? Does it matter which standard errors are used?
 - (ii) Why does the hypothesis $H_0: \beta_{crsgpa} = 1$ make sense? Test this hypothesis against the two-sided alternative at the 5% level, using both standard errors. Describe your conclusions.
 - (iii) Test whether there is an in-season effect on term GPA, using both standard errors. Does the significance level at which the null can be rejected depend on the standard error used?
- 5 The variable *smokes* is a binary variable equal to one if a person smokes, and zero otherwise. Using the data in SMOKE.RAW, we estimate a linear probability model for *smokes*:

$$\begin{aligned} \widehat{smokes} = & .656 - .069 \log(cigpric) + .012 \log(income) - .029 educ \\ & (.855) \quad (.204) \quad \quad (.026) \quad \quad (.006) \\ & [.856] \quad [.207] \quad \quad [.026] \quad \quad [.006] \\ & + .020 age - .00026 age^2 - .101 restaurn - .026 white \\ & (.006) \quad (.00006) \quad (.039) \quad (.052) \\ & [.005] \quad [.00006] \quad [.038] \quad [.050] \\ n = 807, R^2 = .062. \end{aligned}$$

The variable *white* equals one if the respondent is white, and zero otherwise; the other independent variables are defined in Example 8.7. Both the usual and heteroskedasticity-robust standard errors are reported.

- (i) Are there any important differences between the two sets of standard errors?
 - (ii) Holding other factors fixed, if education increases by four years, what happens to the estimated probability of smoking?
 - (iii) At what point does another year of age reduce the probability of smoking?
 - (iv) Interpret the coefficient on the binary variable *restaurn* (a dummy variable equal to one if the person lives in a state with restaurant smoking restrictions).
 - (v) Person number 206 in the data set has the following characteristics: *cigpric* = 67.44, *income* = 6,500, *educ* = 16, *age* = 77, *restaurn* = 0, *white* = 0, and *smokes* = 0. Compute the predicted probability of smoking for this person and comment on the result.
- 6 There are different ways to combine features of the Breusch-Pagan and White tests for heteroskedasticity. One possibility not covered in the text is to run the regression

$$\hat{u}_i^2 \text{ on } x_{i1}, x_{i2}, \dots, x_{ik}, \hat{y}_i^2, i = 1, \dots, n,$$

where the $\hat{u}_i$ are the OLS residuals and the $\hat{y}_i$ are the OLS fitted values. Then, we would test joint significance of $x_{i1}, x_{i2}, \dots, x_{ik}$ and $\hat{y}_i^2$. (Of course, we always include an intercept in this regression.)

- (i) What are the *df* associated with the proposed *F* test for heteroskedasticity?
- (ii) Explain why the *R*-squared from the regression above will always be at least as large as the *R*-squareds for the BP regression and the special case of the White test.
- (iii) Does part (ii) imply that the new test always delivers a smaller *p*-value than either the BP or special case of the White statistic? Explain.

- (iv) Suppose someone suggests also adding $\hat{y}_i$ to the newly proposed test. What do you think of this idea?

7 Consider a model at the employee level,

$$y_{i,e} = \beta_0 + \beta_1 x_{i,e,1} + \beta_2 x_{i,e,2} + \dots + \beta_k x_{i,e,k} + f_i + v_{i,e},$$

where the unobserved variable f_i is a “firm effect” to each employee at a given firm i . The error term $v_{i,e}$ is specific to employee e at firm i . The *composite error* is $u_{i,e} = f_i + v_{i,e}$, such as in equation (8.28).

- Assume that $\text{Var}(f_i) = \sigma_f^2$, $\text{Var}(v_{i,e}) = \sigma_v^2$, and f_i and $v_{i,e}$ are uncorrelated. Show that $\text{Var}(u_{i,e}) = \sigma_f^2 + \sigma_v^2$; call this σ^2 .
- Now suppose that for $e \neq g$, $v_{i,e}$ and $v_{i,g}$ are uncorrelated. Show that $\text{Cov}(u_{i,e}, u_{i,g}) = \sigma_f^2$.
- Let $\bar{u}_i = m_i^{-1} \sum_{e=1}^{m_i} u_{i,e}$ be the average of the composite errors within a firm. Show that $\text{Var}(\bar{u}_i) = \sigma_f^2 + \sigma_v^2/m_i$.
- Discuss the relevance of part (iii) for WLS estimation using data averaged at the firm level, where the weight used for observation i is the usual firm size.

Computer Exercises

C1 Consider the following model to explain sleeping behavior:

$$\text{sleep} = \beta_0 + \beta_1 \text{totwrk} + \beta_2 \text{educ} + \beta_3 \text{age} + \beta_4 \text{age}^2 + \beta_5 \text{yngkid} + \beta_6 \text{male} + u.$$

- Write down a model that allows the variance of u to differ between men and women. The variance should not depend on other factors.
 - Use the data in SLEEP75.RAW to estimate the parameters of the model for heteroskedasticity. (You have to estimate the *sleep* equation by OLS, first, to obtain the OLS residuals.) Is the estimated variance of u higher for men or for women?
 - Is the variance of u statistically different for men and for women?
- C2
- Use the data in HPRICE1.RAW to obtain the heteroskedasticity-robust standard errors for equation (8.17). Discuss any important differences with the usual standard errors.
 - Repeat part (i) for equation (8.18).
 - What does this example suggest about heteroskedasticity and the transformation used for the dependent variable?
- C3 Apply the full White test for heteroskedasticity [see equation (8.19)] to equation (8.18). Using the chi-square form of the statistic, obtain the p -value. What do you conclude?
- C4 Use VOTE1.RAW for this exercise.
- Estimate a model with *voteA* as the dependent variable and *prtystrA*, *democA*, $\log(\text{expendA})$, and $\log(\text{expendB})$ as independent variables. Obtain the OLS residuals, $\hat{u}_i$, and regress these on all of the independent variables. Explain why you obtain $R^2 = 0$.
 - Now, compute the Breusch-Pagan test for heteroskedasticity. Use the F statistic version and report the p -value.
 - Compute the special case of the White test for heteroskedasticity, again using the F statistic form. How strong is the evidence for heteroskedasticity now?

C5 Use the data in PNTSPRD.RAW for this exercise.

- (i) The variable *sprdcvr* is a binary variable equal to one if the Las Vegas point spread for a college basketball game was covered. The expected value of *sprdcvr*, say μ , is the probability that the spread is covered in a randomly selected game. Test $H_0: \mu = .5$ against $H_1: \mu \neq .5$ at the 10% significance level and discuss your findings. (*Hint:* This is easily done using a t test by regressing *sprdcvr* on an intercept only.)
- (ii) How many games in the sample of 553 were played on a neutral court?
- (iii) Estimate the linear probability model

$$\text{sprdcvr} = \beta_0 + \beta_1 \text{favhome} + \beta_2 \text{neutral} + \beta_3 \text{fav25} + \beta_4 \text{und25} + u$$

and report the results in the usual form. (Report the usual OLS standard errors and the heteroskedasticity-robust standard errors.) Which variable is most significant, both practically and statistically?

- (iv) Explain why, under the null hypothesis $H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$, there is no heteroskedasticity in the model.
- (v) Use the usual F statistic to test the hypothesis in part (iv). What do you conclude?
- (vi) Given the previous analysis, would you say that it is possible to systematically predict whether the Las Vegas spread will be covered using information available prior to the game?

C6 In Example 7.12, we estimated a linear probability model for whether a young man was arrested during 1986:

$$\text{arr86} = \beta_0 + \beta_1 \text{pcnv} + \beta_2 \text{avgse} + \beta_3 \text{tottime} + \beta_4 \text{ptime86} + \beta_5 \text{qemp86} + u.$$

- (i) Using the data in CRIME1.RAW, estimate this model by OLS and verify that all fitted values are strictly between zero and one. What are the smallest and largest fitted values?
- (ii) Estimate the equation by weighted least squares, as discussed in Section 8.5.
- (iii) Use the WLS estimates to determine whether *avgse* and *tottime* are jointly significant at the 5% level.

C7 Use the data in LOANAPP.RAW for this exercise.

- (i) Estimate the equation in part (iii) of Computer Exercise C8 in Chapter 7, computing the heteroskedasticity-robust standard errors. Compare the 95% confidence interval on β_{white} with the nonrobust confidence interval.
- (ii) Obtain the fitted values from the regression in part (i). Are any of them less than zero? Are any of them greater than one? What does this mean about applying weighted least squares?

C8 Use the data set GPA1.RAW for this exercise.

- (i) Use OLS to estimate a model relating *colGPA* to *hsGPA*, *ACT*, *skipped*, and *PC*. Obtain the OLS residuals.
- (ii) Compute the special case of the White test for heteroskedasticity. In the regression of $\hat{u}_i^2$ on *colGPA*_{*i*}, *colGPA*_{*i*}², obtain the fitted values, say $\hat{h}_i$.
- (iii) Verify that the fitted values from part (ii) are all strictly positive. Then, obtain the weighted least squares estimates using weights $1/\hat{h}_i$. Compare the weighted least squares estimates for the effect of skipping lectures and the effect of PC ownership with the corresponding OLS estimates. What about their statistical significance?

- (iv) In the WLS estimation from part (iii), obtain heteroskedasticity-robust standard errors. In other words, allow for the fact that the variance function estimated in part (ii) might be misspecified. (See Question 8.4.) Do the standard errors change much from part (iii)?

C9 In Example 8.7, we computed the OLS and a set of WLS estimates in a cigarette demand equation.

- (i) Obtain the OLS estimates in equation (8.35).
- (ii) Obtain the $\hat{h}_i$ used in the WLS estimation of equation (8.36) and reproduce equation (8.36). From this equation, obtain the *unweighted* residuals and fitted values; call these $\hat{u}_i$ and $\hat{y}_i$, respectively. (For example, in Stata, the unweighted residuals and fitted values are given by default.)
- (iii) Let $\check{u}_i = \hat{u}_i/\sqrt{\hat{h}_i}$ and $\check{y}_i = \hat{y}_i/\sqrt{\hat{h}_i}$ be the weighted quantities. Carry out the special case of the White test for heteroskedasticity by regressing $\check{u}_i^2$ on $\check{y}_i, \check{y}_i^2$, being sure to include an intercept, as always. Do you find heteroskedasticity in the weighted residuals?
- (iv) What does the finding from part (iii) imply about the proposed form of heteroskedasticity used in obtaining (8.36)?
- (v) Obtain valid standard errors for the WLS estimates that allow the variance function to be misspecified.

C10 Use the data set 401KSUBS.RAW for this exercise.

- (i) Using OLS, estimate a linear probability model for *e401k*, using as explanatory variables *inc*, *inc*², *age*, *age*², and *male*. Obtain both the usual OLS standard errors and the heteroskedasticity-robust versions. Are there any important differences?
- (ii) In the special case of the White test for heteroskedasticity, where we regress the squared OLS residuals on a quadratic in the OLS fitted values, $\hat{u}_i^2$ on $\hat{y}_i, \hat{y}_i^2$, $i = 1, \dots, n$, argue that the probability limit of the coefficient on $\hat{y}_i$ should be one, the probability limit of the coefficient on $\hat{y}_i^2$ should be -1 , and the probability limit of the intercept should be zero. {Hint: Remember that $\text{Var}(y|x_1, \dots, x_k) = p(\mathbf{x})[1 - p(\mathbf{x})]$, where $p(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$.}
- (iii) For the model estimated from part (i), obtain the White test and see if the coefficient estimates roughly correspond to the theoretical values described in part (ii).
- (iv) After verifying that the fitted values from part (i) are all between zero and one, obtain the weighted least squares estimates of the linear probability model. Do they differ in important ways from the OLS estimates?

C11 Use the data in 401KSUBS.RAW for this question, restricting the sample to *fsize* = 1.

- (i) To the model estimated in Table 8.1, add the interaction term, *e401k* · *inc*. Estimate the equation by OLS and obtain the usual and robust standard errors. What do you conclude about the statistical significance of the interaction term?
- (ii) Now estimate the more general model by WLS using the same weights, *1/inc*, as in Table 8.1. Compute the usual and robust standard error for the WLS estimator. Is the interaction term statistically significant using the robust standard error?
- (iii) Discuss the WLS coefficient on *e401k* in the more general model. Is it of much interest by itself? Explain.

- (iv) Reestimate the model by WLS but use the interaction term $e401k \cdot (inc - 30)$; the average income in the sample is about 29.44. Now interpret the coefficient on $e401k$.

C12 Use the data in MEAP00.RAW to answer this question.

- (i) Estimate the model

$$math4 = \beta_0 + \beta_1 lunch + \beta_2 \log(enroll) + \beta_3 \log(exppp) + u$$

by OLS and obtain the usual standard errors and the fully robust standard errors. How do they generally compare?

- (ii) Apply the special case of the White test for heteroskedasticity. What is the value of the F test? What do you conclude?
- (iii) Obtain $\hat{g}_i$ as the fitted values from the regression $\log(\hat{u}_i^2)$ on $\widehat{math4}_i$, $\widehat{math4}_i^2$, where $\widehat{math4}_i$ are the OLS fitted values and the $\hat{u}_i$ are the OLS residuals. Let $\hat{h}_i = \exp(\hat{g}_i)$. Use the $\hat{h}_i$ to obtain WLS estimates. Are there big differences with the OLS coefficients?
- (iv) Obtain the standard errors for WLS that allow misspecification of the variance function. Do these differ much from the usual WLS standard errors?
- (v) For estimating the effect of spending on $math4$, does OLS or WLS appear to be more precise?

C13 Use the data in FERTIL2.RAW to answer this question.

- (i) Estimate the model

$$children = \beta_0 + \beta_1 age + \beta_2 age^2 + \beta_3 educ + \beta_4 electric + \beta_5 urban + u$$

and report the usual and heteroskedasticity-robust standard errors. Are the robust standard errors always bigger than the nonrobust ones?

- (ii) Add the three religious dummy variables and test whether they are jointly significant. What are the p -values for the nonrobust and robust tests?
- (iii) From the regression in part (ii), obtain the fitted values $\hat{y}$ and the residuals, $\hat{u}$. Regress $\hat{u}^2$ on $\hat{y}$, $\hat{y}^2$ and test the joint significance of the two regressors. Conclude that heteroskedasticity is present in the equation for *children*.
- (iv) Would you say the heteroskedasticity you found in part (iii) is practically important?

C14 Use the data in BEAUTY.RAW for this question.

- (i) Using the data pooled for men and women, estimate the equation

$$lwage = \beta_0 + \beta_1 belavg + \beta_2 abvavg + \beta_3 female + \beta_4 educ + \beta_5 exper + \beta_5 exper^2 + u,$$

and report the results using heteroskedasticity-robust standard errors below coefficients. Are any of the coefficients surprising in either their signs or magnitudes? Is the coefficient on *female* practically large and statistically significant?

- (ii) Add interactions of *female* with all other explanatory variables in the equation from part (i) (five interactions in all). Compute the usual F test of joint significance of the five interactions and a heteroskedasticity-robust version. Does using the heteroskedasticity-robust version change the outcome in any important way?
- (iii) In the full model with interactions, determine whether those involving the looks variables—*female* • *belavg* and *female* • *abvavg*—are jointly significant. Are their coefficients practically small?